Unified Terpene Pharmacology: Bridging Aromatherapy, Cannabis Science, and Forest Medicine

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Abbreviations: A2a, adenosine A2a receptor; AChE, acetylcholinesterase; BCP, β-caryophyllene; BDNF, brain-derived neurotrophic factor; CB1/CB2, cannabinoid receptor type 1/2; CBD, cannabidiol; CI, Combination Index; CSA, Controlled Substances Act; DCC, Department of Cannabis Control; DEA, Drug Enforcement Administration; DML, Designed Multiple Ligand; ERK, extracellular signal-regulated kinase; ES, effect size; FDA, Food and Drug Administration; GABA-A, gamma-aminobutyric acid type A receptor; GRAS, Generally Recognized as Safe; IND, Investigational New Drug; NCCIH, National Center for Complementary and Integrative Health; NIDA, National Institute on Drug Abuse; NIH, National Institutes of Health; NK, natural killer; NF-κB, nuclear factor kappa B; PK, pharmacokinetics; RCT, randomized controlled trial; THC, Δ9-tetrahydrocannabinol; TRP, transient receptor potential.

Significance Statement

This review is the first comprehensive integration of three parallel pharmacological literatures—cannabis terpene research, clinical aromatherapy, and forest bathing immunology—that have studied identical molecules (linalool, α-pinene, d-limonene, β-caryophyllene) in disciplinary isolation since 1937. The 2024 identification of the adenosine A2a receptor as the primary terpene analgesic mechanism, combined with the characterization of A2a-CB1 receptor heteromers as the molecular substrate for terpene-cannabinoid crosstalk, provides the mechanistic linchpin for reunification. By assembling evidence from over 100 primary sources across these three fields—including the first controlled human demonstration of a specific terpene-cannabinoid interaction (selective d-limonene-mediated anxiolysis during THC administration)—this review establishes a unified terpene pharmacology framework with direct translational implications for pain management, anxiety treatment, and rational phytopharmaceutical design.

Abstract

Terpenes constitute the largest class of plant secondary metabolites and are the primary determinants of cannabis chemotype diversity, yet their pharmacology has been fragmented across three parallel disciplines—cannabis science, clinical aromatherapy, and forest medicine—since the coincidental publication of both the Marihuana Tax Act and Gattefossé's Aromathérapie in 1937. This review systematically integrates evidence from all three fields to establish a unified terpene pharmacology framework.

We demonstrate that seven major terpenes engage at least seven distinct molecular targets extending far beyond the cannabinoid CB1/CB2 system, including adenosine A2a receptors, GABA-A receptors, TRP ion channels, serotonergic pathways, cholinergic enzymes, and NF-κB inflammatory cascades. The 2024 discovery that terpenes produce morphine-equivalent analgesia

through the adenosine A2a receptor (Schwarz et al., PAIN) resolves the apparent Finlay et al. (2020) null result that had been interpreted as disproving the entourage effect. The A2a-CB1 heteromer system, mapped to striatopallidal neurons and hippocampal CA1 synapses, provides the molecular substrate for terpene-cannabinoid cross-talk with brain region–specific functional consequences.

Clinical evidence spans 15,000–25,000 aromatherapy RCT participants, the Silexan linalool program achieving paroxetine-equivalent anxiolysis in 1,213 double-blind RCT subjects, forest bathing immunology demonstrating very large NK cell activation effects (ES = 2.50), and the first controlled human terpene-cannabinoid interaction study showing selective d-limonene-mediated anxiolysis at 15 mg vaporized dose (Spindle et al., 2024).

Network pharmacology provides the intellectual framework for understanding multi-target terpene activity as an evolved design feature amenable to rational optimization through cross-botanical blending. Three critical unresolved issues are identified: (1) the 3–4 order-of-magnitude dose gap between preclinical and human terpene exposure; (2) failure of terpene inhalation to produce antinociception in the primary mouse model; and (3) existential threat to the field’s primary federal funding sponsor (NCCIH). The pre-loading paradigm—using essential oil-derived terpene inhalation to prime A2a receptors before cannabis consumption—is proposed as the framework’s most direct translational application.

I. Terpene Receptor Pharmacology Reveals a Multi-Target Landscape Far Beyond CB1/CB2

The pharmacology of plant-derived terpenoids has historically been organized around individual receptor targets: β-caryophyllene as a CB2 agonist, linalool as a GABA-A modulator, limonene as a serotonergic agent. This single-target framing, inherited from the rational drug design paradigm that dominated late twentieth-century pharmacology, fundamentally misrepresents the polypharmacological reality of terpene action. Each major cannabis terpene engages multiple receptor systems at pharmacologically relevant concentrations, and the recent identification of adenosine A2a receptors as a primary mediator of terpene antinociception (Schwarz et al., 2024) has reframed the entire field. This section provides a systematic receptor-by-receptor analysis of the seven most abundant cannabis terpenes, integrating data from cannabis pharmacology, aromatherapy research, and forest medicine—three disciplines that have independently characterized the same molecular targets without meaningful cross-citation.

A. β-Caryophyllene: The Dietary Cannabinoid and Its Expanding Target Profile

β-Caryophyllene (BCP) occupies a unique position in terpene pharmacology as the only terpene with unambiguous, high-affinity binding to a cannabinoid receptor. Gertsch et al. (2008) identified (E)-BCP as a full, selective agonist at the human CB2 receptor using radioligand displacement of [^{3}H]CP55,940 in HEK293 cells, reporting Ki values of 155 ± 4 nM for the (E)-isomer and 485 ± 36 nM for the (Z)-isomer—affinities comparable to those of the endocannabinoid 2-arachidonoylglycerol. Critically, BCP displays no measurable CB1 binding, making it entirely CB2-selective and eliminating any psychoactive liability through this pathway.

Downstream signaling proceeds through canonical Gi/Go coupling, with dose-dependent inhibition of forskolin-stimulated adenylate cyclase ($R^2 > 0.97$), intracellular calcium transients in CB2-expressing HL60 cells ($R^2 = 0.99$, abolished by the CB2 antagonist SR144528 at 1 μM), and activation of ERK1/2 and p38 MAPK in primary human monocytes. The functional

consequence of this signaling cascade is robust anti-inflammatory activity: BCP at 500 nM inhibits LPS-stimulated IL-1β and TNF-α expression in human peripheral whole blood, effects blocked by the CB2 antagonist AM630. The in vivo confirmation came from knockout studies demonstrating that oral BCP at 5 mg/kg strongly reduces carrageenan-induced paw edema in wild-type mice but produces no effect in CB2 knockout ($Cnr2^{-/-}$) animals, establishing CB2-dependent anti-inflammatory action with genetic certainty.

Recent work (2023–2025) has extended BCP's mechanistic profile beyond CB2 alone. In diabetic cardiomyopathy models, BCP suppresses the TLR4/NF-κB/MAPK signaling cascade (ACS Pharmacol. Transl. Sci., 2023), with effects abrogated by AM630—confirming CB2 dependence in a disease-relevant context. In LPS-stimulated HaCaT keratinocytes, BCP at 10 μM reduces phospho-NF-κB p65, COX-2, and IL-1β, with effects reversed by AM630 at 100 nM (Pharmaceuticals, 2024). Particularly noteworthy for the cross-botanical framework is the demonstration that BCP combined with cannabidiol (CBD) produces synergistic cytokine suppression via dual CB2/PPARγ activation (Pharmaceuticals, 2024), establishing a phytocannabinoid-terpenoid interaction with defined molecular targets. Additional receptors engaged by BCP include TRPV1 and PPARγ, though the relative contribution of these targets to BCP's therapeutic profile remains to be quantified. Despite this substantial preclinical literature, no large-scale Phase II/III clinical trials of purified BCP have been reported as of early 2026, representing a significant translational gap for the most pharmacologically characterized cannabis terpene.

The significance of BCP for the unified framework extends beyond its receptor pharmacology. BCP is a common dietary constituent—present in black pepper, cloves, rosemary, oregano, and hops—and received explicit Generally Recognized as Safe (GRAS) status from the FDA. It is simultaneously one of the most abundant terpenes in cannabis flower, appearing in the majority of commercially available strains. The same molecule that cannabis researchers characterize as a "phytocannabinoid" is freely sold in essential oil formulations and consumed

daily as a food additive, a regulatory paradox that encapsulates the disciplinary fragmentation this review seeks to address.

B. d-Limonene: From Citrus Monoterpene to A2a Receptor Agonist

The pharmacological identity of d-limonene has undergone a fundamental revision. Long classified as a mood-elevating agent with poorly defined mechanisms, limonene's story was transformed by two lines of evidence: the identification of adenosine A2a receptor agonism (Park et al., 2011) and the demonstration of selective cannabinoid anxiolysis in humans (Spindle et al., 2024). Together, these findings position limonene at the nexus of the terpene-cannabinoid interaction and illustrate why a unified framework across disciplines is not merely intellectually satisfying but pharmacologically necessary.

Park et al. (2011) demonstrated that limonene exhibits selective affinity for adenosine A2a receptors in rat brain cerebral cortex radioligand binding assays. In CHO cells transfected with human A2a, limonene increased cytosolic cAMP, activated protein kinase A (PKA), phosphorylated CREB, and elevated cytosolic calcium—the complete canonical A2a signaling cascade. Although quantitative Ki values were not reported in this initial study, selective displacement was confirmed. This finding was extended by Siddiquee et al. (2021), who demonstrated limonene-induced A2a (and possibly A2b) receptor activation sufficient to reduce airway inflammation in an ovalbumin-sensitized mouse asthma model, with effects absent in A2a knockout mice—providing the genetic confirmation of receptor dependence. Park et al. (2021) further demonstrated limonene's anti-anxiety activity at 5–10 mg/kg IP in C57BL/6N mice, an effect blocked by the A2a antagonist SCH58261 and mediated through modulation of dopaminergic and GABAergic neuronal function in the striatum.

The clinical significance of limonene's A2a agonism is amplified by its intersection with the cannabinoid system. The Spindle et al. (2024) study at Johns Hopkins represents the first controlled human evidence for a specific terpene-cannabinoid interaction: 20 healthy adults in a

randomized, double-blind, within-subject crossover design vaporized d-limonene (1, 5, and 15 mg) combined with THC (15 and 30 mg). The results were remarkably selective: 30 mg THC combined with 15 mg d-limonene significantly reduced self-reported ratings of "anxious/nervous" and "paranoid" compared to 30 mg THC alone, following an orderly dose-response pattern. Critically, limonene did not alter any other THC effect—not the subjective "high," not cognitive impairment, not vital signs, and not THC pharmacokinetics. Limonene alone produced no measurable pharmacodynamic effects. This selective anxiolysis, occurring without modification of the desired therapeutic or recreational effects of THC, represents a clinically actionable finding with immediate translational implications.

The mechanism underlying this selective interaction likely involves the A2a-CB1 receptor heteromer system (discussed in detail in Section VI), wherein A2a activation modulates CB1 signaling in a brain region–specific manner. Importantly, however, limonene's anxiolytic effects were not blocked by flumazenil (Ferber et al., 2020), indicating a mechanism distinct from classical benzodiazepine-site GABA-A modulation—consistent with A2a-mediated rather than GABAergic anxiolysis. The aromatherapy literature has documented citrus oil mood effects for decades (Watanabe et al., 2015; Komori et al., 1995), but without the molecular resolution provided by the A2a receptor identification. The forest medicine literature similarly documents limonene as a major "phytoncide" in forest air (Li et al., 2006), with documented immune effects, yet neither field has integrated the A2a mechanism. These three parallel literatures—cannabis pharmacology identifying selective THC anxiolysis, aromatherapy documenting citrus mood effects, and forest medicine characterizing phytoncide immune activity—describe the same molecule acting through the same receptor, a convergence invisible to researchers working within any single discipline.

C. Linalool: GABA-A Modulation, the Olfactory Gate, and the Route-Dependence Problem

Linalool, the dominant monoterpene in lavender essential oil and a significant constituent of many cannabis chemovars, presents a pharmacological case study in route-dependent mechanism. The definitive experiment was provided by Harada et al. (2018), who used an elegant anosmia paradigm to demonstrate that linalool's anxiolytic effects depend entirely on olfactory input. Using 240 male C57BL/6N mice (10/group), they showed that linalool odor dose-dependently increased time in the light box and open arms of the elevated plus maze ($p < 0.01$ to $p < 0.0001$ versus controls). The critical manipulation: destruction of the olfactory epithelium with 3-methylindole (300 mg/kg IP, administered two weeks prior) completely abolished the anxiolytic effect in anosmic mice. This result establishes that linalool's behavioral effects, at least at inhalation concentrations achievable through environmental exposure, require an intact olfactory system and are not produced by systemic absorption acting directly on central receptors.

The downstream receptor target was identified through selective antagonism: anxiolysis was blocked by flumazenil, a benzodiazepine-site antagonist, but not by WAY-100635, a 5-HT1A antagonist ($p = 0.7374$), confirming GABA-A–benzodiazepine receptor mediation through olfactory signaling rather than serotonergic pathways. The electrophysiological basis for this interaction was characterized by Milanos et al. (2017), who used whole-cell patch-clamp recordings in HEK293 cells expressing α1β2γ2 GABA-A receptors to show that linalool at 2 mM enhanced GABA-evoked currents to 160 ± 14% of GABA alone ($p < 0.05$). A subtle but important mechanistic distinction emerged: linalool acts allosterically at a site independent of the γ2 subunit, meaning it does not bind the classical benzodiazepine site between α and γ subunits. This places linalool's modulation at a pharmacologically distinct site from benzodiazepine drugs, despite the flumazenil sensitivity of the behavioral response—suggesting that the olfactory-

limbic pathway may engage the benzodiazepine site indirectly while linalool's direct receptor action occurs elsewhere on the GABA-A complex.

Additional targets extend linalool's pharmacological profile beyond GABA-A. Hashimoto et al. (2023) demonstrated TRPA1 inhibition and voltage-gated Ca^{2+} channel blockade in sensory neurons, providing a mechanistic basis for analgesic effects independent of GABA-A modulation. The glutamatergic system is also engaged: Elisabetsky et al. (1995) and Batista et al. (2008) demonstrated NMDA receptor antagonism, adding an anti-excitotoxic dimension to linalool's neuroprotective profile. This multi-target engagement—GABA-A allosteric modulation, TRPA1 inhibition, calcium channel blockade, and NMDA antagonism—exemplifies the polypharmacological architecture that network pharmacology frameworks (Hopkins, 2008) predict should be therapeutically superior to single-target interventions.

The clinical translation of linalool's pharmacology is most advanced through the Silexan program—a standardized oral lavender oil preparation containing linalool (25–46%) and linalyl acetate (25–57%) that has completed five double-blind, placebo-controlled trials totaling 1,213 participants (Dold et al., 2023). In the pivotal trial (Kasper et al., 2014), 539 adults with generalized anxiety disorder were randomized to Silexan 160 mg/day, Silexan 80 mg/day, paroxetine 20 mg/day, or placebo for 10 weeks. Hamilton Anxiety Scale reductions for Silexan 160 mg reached -14.1 ± 9.3 points ($p < 0.01$ versus placebo), compared to -11.3 ± 8.0 for paroxetine ($p = 0.10$ versus placebo in the full analysis set). Remission rates (HAMA < 10) were 46.3% for high-dose Silexan versus 34.1% for paroxetine, with no withdrawal symptoms—a clinically significant advantage over SSRIs. Silexan is now authorized in Germany as Lasea® and was included in the Bandelow et al. (2023) clinical guidelines as a recommended medication for anxiety. This represents the most advanced clinical translation of any terpene, yet it appears in the psychiatric and phytotherapy literature, essentially invisible to the cannabis pharmacology community that studies the same molecule under different nomenclature.

D. α-Pinene: Cholinergic Enhancement, Neurotrophin Signaling, and the Bioavailability Advantage

α-Pinene is the most abundant terpene in the terrestrial biosphere, the dominant volatile in coniferous forest air, a significant constituent of rosemary essential oil, and a prominent terpene in numerous cannabis chemovars. Its pharmacology spans cholinergic, GABAergic, and neurotrophic pathways, with a notable bioavailability advantage that distinguishes it from most other terpenes under consideration.

Acetylcholinesterase (AChE) inhibition was quantified by Miyazawa and Yamafuji (2005): (+)-α-pinene IC_{50} = 0.40 mM and (−)-α-pinene IC_{50} = 0.44 mM against bovine erythrocyte AChE. These values are orders of magnitude weaker than pharmaceutical AChE inhibitors such as donepezil (IC_{50} approximately 6–23 nM), a comparison that would seem to dismiss α-pinene's cholinergic relevance. However, the pharmacokinetic context fundamentally alters this assessment. Falk and Hagberg (1990) documented α-pinene inhalation pharmacokinetics in eight healthy males, finding approximately 60% pulmonary uptake—a remarkably efficient absorption rate that delivers substantial circulating concentrations from relatively modest environmental exposure. With a distribution half-life of approximately 5 minutes, an elimination half-life of 26–38 minutes, and a third-phase half-life of 695 minutes (approximately 11.6 hours) reflecting high adipose affinity, inhaled α-pinene achieves rapid CNS access with prolonged tissue residence. Direct measurement by Satou et al. (2016) confirmed that α-pinene reaches maximal brain concentrations within 30 minutes of inhalation at levels exceeding 90 μg/g. Notably, brain concentrations were two-fold higher when α-pinene was inhaled as part of a mixed essential oil blend versus single-compound inhalation, suggesting synergistic absorption mechanisms that are directly relevant to the cross-botanical framework.

Beyond cholinergic modulation, α-pinene engages the BDNF/TrkB/CREB neurotrophin signaling pathway, a mechanism with implications for neuroprotection and memory enhancement. Hashemi et al. (2023) demonstrated that α-pinene (50 mg/kg/day IP for 21 days)

reversed kainic acid–induced decreases in hippocampal BDNF, TrkB, phospho-TrkB, CREB, and phospho-CREB in Wistar rats, with corresponding preservation of CA1 pyramidal neurons and recovery of spatial and inhibitory avoidance memory. Kasuya et al. (2015) showed that inhaled α-pinene increases BDNF and tyrosine hydroxylase mRNA in mouse brain, providing evidence that the inhalation route can engage neurotrophin signaling. In Huntington's disease models (2024/2025), α-pinene at 1–10 mg/kg elevated BDNF, TrkA, TrkB, and CREB gene expression while decreasing p75NTR in cortex and hippocampus—a profile that shifts the neurotrophin balance from pro-apoptotic to pro-survival signaling. Additionally, Yang et al. (2016) identified α-pinene as a partial modulator at the GABA-A benzodiazepine binding site that enhances non-REM sleep, adding a sedative-hypnotic dimension relevant to cannabis strains with prominent α-pinene content.

The convergence across disciplines for α-pinene is particularly striking. Forest medicine researchers measure α-pinene at concentrations up to 13.05 μg/m³ in pine forest air, documenting immune enhancement and stress reduction (Li et al., 2008a). Aromatherapy researchers study α-pinene as a component of rosemary essential oil, correlating blood levels with cognitive performance (Moss and Oliver, 2012). Cannabis researchers characterize α-pinene-dominant strains as producing alert, focused effects with memory preservation. All three communities are studying the same molecule's effects on overlapping neuronal targets—AChE inhibition, BDNF signaling, GABA-A modulation—yet the cross-citation rate between these literatures approaches zero.

E. Myrcene: The Paradox of CB1 Engagement Without Direct Agonism

Myrcene, the most abundant terpene in many cannabis chemovars and the compound popularly associated with the sedative "couch lock" effect of indica-type strains, presents a pharmacological paradox that illuminates the complexity of indirect receptor engagement. The classical pharmacology was established by Rao et al. (1990), who demonstrated that myrcene

analgesia at 10–40 mg/kg involves both opioidergic mechanisms (blocked by naloxone) and α2-adrenergic mechanisms (blocked by yohimbine), proposing a model wherein α2-adrenoceptor stimulation triggers release of endogenous opioids. Myrcene also activates TRPV1 in a capsazepine-sensitive manner (Heblinski et al., Channels), adding a nociceptive processing dimension.

The recent finding by Alayoubi et al. (2025) has sharpened the paradox considerably. In CCI neuropathic pain models, myrcene (1–200 mg/kg IP) dose-dependently increased mechanical nociceptive thresholds, with notably greater potency in females. The CB1 antagonist rimonabant blocked this antinociceptive effect—yet the TRUPATH assay confirmed that myrcene does not directly activate CB1 at 10 μM, nor does it alter CB1 signaling by CP55,940, anandamide, or 2-AG. This "engagement without agonism" represents a mechanistic puzzle with several possible resolutions: myrcene may modulate endocannabinoid tone by inhibiting FAAH or MAGL (the primary enzymes responsible for anandamide and 2-AG degradation), alter membrane fluidity in a manner that affects CB1 conformational dynamics, or engage an allosteric site not detected by the TRUPATH assay. Each of these mechanisms would produce CB1-dependent behavioral effects without direct orthosteric agonism, consistent with the observed pharmacology.

The popular attribution of sedative "couch lock" properties to myrcene-dominant cannabis strains deserves critical scrutiny. McKenna et al. (2022) specifically noted no effect of myrcene on locomotion at analgesic doses, and the threshold of 0.5% myrcene content often cited as the indica-sativa demarcation has no controlled experimental basis. The Jikomes et al. (2022) analysis of 89,923 commercial cannabis samples confirmed that terpene profiles—not indica/sativa labels—are the meaningful chemical differentiators, but the specific attribution of sedation to myrcene concentration remains a pharmacological hypothesis awaiting rigorous human testing.

F. 1,8-Cineole, α-Humulene, and Minor Terpenes: Completing the Receptor Landscape

The remaining terpenes that constitute cannabis profiles each contribute distinct pharmacological targets to the overall polypharmacological architecture. 1,8-Cineole (eucalyptol), the principal component of eucalyptus oil and a significant constituent of rosemary, primarily acts as a TRPM8 agonist with an EC_{50} of 120.4 μM at human TRPM8—notably six-fold more potent at the human receptor than the mouse ortholog (Caceres et al., 2017). Cineole simultaneously acts as a TRPA1 antagonist (Takaishi et al., 2012), and TRPM8 knockout abolishes its anti-inflammatory effects (Caceres, 2017), establishing TRP channel modulation as its primary mechanism. The Moss research program (2003–2025) has documented the cognitive enhancement produced by rosemary aroma—of which 1,8-cineole is the principal active constituent—culminating in the 2012 demonstration that serum 1,8-cineole concentrations after rosemary aroma exposure correlate with cognitive performance in healthy volunteers (serial threes accuracy $r = 0.469$, $p = 0.037$; serial threes reaction time $r = -0.502$, $p = 0.024$), with medium-to-large effect sizes ($r^2 = 0.19–0.25$). This remains the only published demonstration of a direct correlation between circulating terpene levels and cognitive outcomes in humans.

α-Humulene shares sesquiterpene ancestry with BCP, differing by a single ring structure, but this structural distinction has profound pharmacological consequences: humulene lacks CB2 affinity (Ki > 20 μM; Gertsch, 2008). Its anti-inflammatory action instead proceeds through NF-κB and AP-1 inhibition, producing dose-dependent reductions in TNF-α, IL-1β, PGE_2, iNOS, and COX-2 at 50 mg/kg oral (Fernandes and Medeiros, 2007). Among the less abundant terpenes, geraniol merits particular attention as the most pharmacologically characterized: it acts as a potent TRPA1 modulator (EC_{50}/IC_{50} < 1 μM for derivatives; Ortar et al., 2014) and was the most effective terpene in the Schwarz et al. (2024) chemotherapy-induced peripheral neuropathy model. α-Bisabolol inhibits NF-κB via IκBα phosphorylation blockade and activates PPARγ (Kim et al., 2011). Notably, terpinolene and β-ocimene—both significant contributors to the

terpene profiles of popular cannabis strains—remain pharmacologically poorly characterized, with no specific G protein–coupled receptor binding data published for either compound. This gap is not trivial: terpinolene-dominant strains represent a distinct chemotype in the Jikomes classification, yet their pharmacological basis remains essentially undefined.

G. TRP Channel Interactions: Correcting a Pervasive Mis-Citation

A methodological note warranting emphasis concerns the frequently cited De Petrocellis et al. (2011) study in the British Journal of Pharmacology. This paper is routinely referenced in reviews of terpene pharmacology as evidence for terpene–TRP channel interactions. In fact, it tested phytocannabinoids (CBD, THC, CBC, CBG, and others) against TRP channels, not terpenes. The key findings—CBC activating TRPA1 at EC_{50} = 90 nM, CBD antagonizing TRPM8 at IC_{50} = 60 nM, and CBD activating TRPV1 at EC_{50} = 1–3.5 μM—are cannabinoid-TRP data that, while relevant to the broader entourage effect framework, do not constitute direct evidence for terpene–TRP interactions. For actual terpene–TRP data, the primary evidence comes from Caceres (2017) for 1,8-cineole at TRPM8, Takaishi et al. (2012) for cineole at TRPA1, and Heblinski et al. for myrcene at TRPV1. This citation error, propagated across multiple review articles, illustrates a broader quality concern in the terpene pharmacology literature that the present review attempts to address through systematic source verification.

H. The Finlay Null Result and Its Resolution Through A2a Pharmacology

The interpretation of the "entourage effect" pivoted on a single negative result that, for several years, appeared to undermine the hypothesis entirely. Finlay et al. (2020) tested α-pinene, β-pinene, β-caryophyllene, linalool, and limonene individually and in mixtures at concentrations up to 10 μM in HEK293 cells expressing human CB1 or CB2. No terpene displaced [^{3}H]CP55,940, altered THC or CBD binding affinity, or modulated cAMP signaling. This was interpreted as definitive evidence against a CB1/CB2-mediated entourage effect, and it cast a long shadow over the field.

However, three subsequent lines of evidence have reframed the Finlay result as technically correct but directed at the wrong question. First, LaVigne et al. (2021) demonstrated CB1-dependent ERK activation by terpenes at higher concentrations using more sensitive functional assays, suggesting that the Finlay study's concentration range and endpoint selection may have been insufficient to detect allosteric or pathway-biased activity. Second, Raz et al. (2023) detected all 16 tested cannabis terpenes activating CB1 at 10–50% of THC efficacy using the highly sensitive GIRK channel assay in Xenopus oocytes—an assay platform with superior signal-to-noise compared to the radioligand displacement and cAMP assays employed by Finlay. Third, and most decisively, Schwarz et al. (2024) demonstrated that terpene antinociception is primarily mediated via adenosine A2a receptors, not cannabinoid receptors.

The resolution is conceptually elegant: the Finlay result confirmed that terpenes show negligible direct CB1/CB2 orthosteric binding, but the question it answered—whether terpenes modulate cannabinoid effects through direct cannabinoid receptor activation—was not the only mechanism through which terpene-cannabinoid synergy could operate. The A2a receptor provides an independent analgesic pathway through which terpenes produce antinociception roughly equivalent to 10 mg/kg morphine (Schwarz et al., 2024), while allosteric CB1 modulation detectable only with sensitive functional assays may contribute secondarily (Raz et al., 2023). The entourage effect, properly understood, does not require direct cannabinoid receptor activation by terpenes; A2a provides the primary mechanism, the A2a-CB1 heteromer system provides the molecular substrate for crosstalk, and multiple additional receptor targets (GABA-A, TRP channels, opioidergic, adrenergic) contribute to the overall polypharmacological profile.

I. Section Summary: A Polypharmacological Architecture Demanding Unified Analysis

Table 1 summarizes the receptor targets, binding affinities, functional consequences, and key references for each major terpene. Several features of this landscape merit emphasis. First, no terpene is a single-target agent: even β-caryophyllene, the most selectively characterized, engages CB2, TRPV1, PPARγ, and the TLR4/NF-κB axis. Second, the dose ranges producing receptor engagement vary by orders of magnitude—from nanomolar CB2 binding by BCP to millimolar AChE inhibition by α-pinene—making bioavailability and route of administration critical determinants of which targets are engaged in any given exposure scenario. Third, the 2024 identification of A2a as a primary analgesic mechanism (Schwarz et al., 2024) does not replace the previously characterized targets but adds a new layer to an already complex polypharmacological architecture. Fourth, two terpenes that define distinct cannabis chemotypes—terpinolene and β-ocimene—lack any published GPCR binding data, representing a critical gap in the evidence base.

The receptor pharmacology reviewed here was established independently across cannabis science, aromatherapy, and forest medicine. The Gertsch CB2 data were published in a general science journal (PNAS), the Harada olfactory gate study in a behavioral neuroscience journal, the Li phytoncide data in immunopharmacology journals, and the Schwarz A2a breakthrough in a pain specialty journal. No single discipline commanded the full picture. Reunifying these data streams under a common pharmacological framework is not merely an exercise in interdisciplinary citation—it is a prerequisite for understanding how terpenes actually work in the complex, multi-source exposure scenarios that characterize both therapeutic and recreational use.

Table 1. Summary of receptor targets, binding affinities, functional consequences, and key references for major cannabis terpenes. BDZ, benzodiazepine; CIPN, chemotherapy-induced

peripheral neuropathy; AChE, acetylcholinesterase; BuChE, butyrylcholinesterase; BBB, blood-brain barrier. Dash (—) indicates no published data.

Terpene	Primary Receptor Target(s)	Binding Affinity / Potency	Functional Consequence	Key Reference(s)
β-Caryophyllene	CB2 (full agonist); TLR4/NF-κB; PPARγ; TRPV1	CB2 Ki = 155 ± 4 nM (E-isomer); 485 ± 36 nM (Z-isomer)	Anti-inflammatory (IL-1β, TNF-α suppression); CB2-dependent analgesia; synergy with CBD via CB2/PPARγ	Gertsch et al. 2008; Pharmaceuticals 2024
d-Limonene	Adenosine A2a (agonist); possibly A2b	Selective A2a displacement confirmed; Ki not quantified	Anxiolysis (A2a-dependent, blocked by SCH58261); selective attenuation of THC-induced anxiety in humans	Park et al. 2011, 2021; Siddiquee et al. 2021; Spindle et al. 2024
Linalool	GABA-A (allosteric, non-BDZ site); TRPA1; voltage-gated Ca^{2+} channels; NMDA	GABA-A: 160 ± 14% enhancement at 2 mM	Anxiolysis (olfactory-dependent, flumazenil-sensitive); analgesic; anti-excitotoxic	Harada et al. 2018; Milanos et al. 2017; Hashimoto et al. 2023
α-Pinene	AChE (inhibitor); BDNF/TrkB/CREB pathway; GABA-A (BDZ partial modulator)	AChE IC_{50} = 0.40 mM (+); 0.44 mM (−)	Cognitive enhancement; neurotrophin signaling; non-REM sleep promotion; ~60% pulmonary bioavailability	Miyazawa & Yamafuji 2005; Hashemi et al. 2023; Yang et al. 2016
Myrcene	α2-Adrenergic; opioidergic (indirect); TRPV1; CB1 (indirect, engagement without agonism)	CB1 antagonist-sensitive but no direct CB1 activation at 10 μM		

Analgesia (naloxone- and yohimbine-blocked); greater potency in females; "couch lock" unconfirmed Rao et al. 1990; Alayoubi et al. 2025; McKenna et al. 2022

1,8-Cineole TRPM8 (agonist); TRPA1 (antagonist); AChE/BuChE TRPM8 EC_{50} = 120.4 μM (human, 6× > mouse) Cognitive enhancement (correlated with serum levels); anti-inflammatory (36% steroid sparing); BBB opening Caceres et al. 2017; Moss & Oliver 2012; Juergens et al. 2003

α-Humulene NF-κB/AP-1 (inhibitor); no CB2 affinity CB2 Ki > 20 μM Anti-inflammatory (TNF-α, IL-1β, PGE_2, iNOS, COX-2 reduction) Gertsch 2008; Fernandes & Medeiros 2007

Geraniol TRPA1 (potent modulator); A2a (antinociceptive) TRPA1 EC_{50}/IC_{50} < 1 μM (derivatives) Most effective terpene in CIPN pain model; A2a-mediated antinociception Ortar et al. 2014; Schwarz et al. 2024

α-Bisabolol NF-κB (IκBα blockade); PPARγ (activator) Not quantified Anti-inflammatory; anxiolytic in preclinical models Kim et al. 2011; Jenkins et al. 2023

Terpinolene No GPCR data published — Pharmacologically uncharacterized despite defining a distinct cannabis chemotype —

β-Ocimene No GPCR data published — Pharmacologically uncharacterized —

Section 1 word count: ~4,800 words (of target 15,000–25,000 total manuscript)

Table 1. Summary of receptor targets, binding affinities, functional consequences, and key references for major cannabis terpenes. BDZ, benzodiazepine; CIPN, chemotherapy-induced peripheral neuropathy; AChE, acetylcholinesterase; BuChE, butyrylcholinesterase; BBB, blood-brain barrier. Dash (—) indicates no published data.

Terpene	Primary Receptor Target(s)	Binding Affinity / Potency	Functional Consequence	Key Reference(s)
β-Caryophyllene	CB2 (full agonist); TLR4/NF-κB; PPARγ; TRPV1	CB2 Ki = 155 ± 4 nM (E-isomer); 485 ± 36 nM (Z-isomer)	Anti-inflammatory (IL-1β, TNF-α suppression); CB2-dependent analgesia; synergy with CBD via CB2/PPARγ	Gertsch et al. 2008; Pharmaceuticals 2024
d-Limonene	Adenosine A2a (agonist); possibly A2b	Selective A2a displacement confirmed; Ki not quantified	Anxiolysis (A2a-dependent, blocked by SCH58261); selective attenuation of THC-induced anxiety in humans	Park et al. 2011, 2021; Siddiquee et al. 2021; Spindle et al. 2024
Linalool	GABA-A (allosteric, non-BDZ site); TRPA1; voltage-gated Ca^{2+} channels; NMDA	GABA-A: 160 ± 14% enhancement at 2 mM	Anxiolysis (olfactory-dependent, flumazenil-sensitive); analgesic; anti-excitotoxic	Harada et al. 2018; Milanos et al. 2017; Hashimoto et al. 2023
α-Pinene	AChE (inhibitor); BDNF/TrkB/CREB pathway; GABA-A (BDZ partial modulator)	AChE IC_{50} = 0.40 mM (+); 0.44 mM (−)	Cognitive enhancement; neurotrophin signaling; non-REM sleep promotion; ~60% pulmonary bioavailability	Miyazawa & Yamafuji 2005; Hashemi et al. 2023; Yang et al. 2016

Myrcene	α2-Adrenergic; opioidergic (indirect); TRPV1; CB1 (indirect, engagement without agonism)	CB1 antagonist-sensitive but no direct CB1 activation at 10 μM	Analgesia (naloxone- and yohimbine-blocked); greater potency in females; "couch lock" unconfirmed	Rao et al. 1990; Alayoubi et al. 2025; McKenna et al. 2022
1,8-Cineole	TRPM8 (agonist); TRPA1 (antagonist); AChE/BuChE	TRPM8 EC_{50} = 120.4 μM (human, 6× > mouse)	Cognitive enhancement (correlated with serum levels); anti-inflammatory (36% steroid sparing); BBB opening	Caceres et al. 2017; Moss & Oliver 2012; Juergens et al. 2003
α-Humulene	NF-κB/AP-1 (inhibitor); no CB2 affinity	CB2 Ki > 20 μM	Anti-inflammatory (TNF-α, IL-1β, PGE_2, iNOS, COX-2 reduction)	Gertsch 2008; Fernandes & Medeiros 2007
Geraniol	TRPA1 (potent modulator); A2a (antinociceptive)	TRPA1 EC_{50}/IC_{50} < 1 μM (derivatives)	Most effective terpene in CIPN pain model; A2a-mediated antinociception	Ortar et al. 2014; Schwarz et al. 2024
α-Bisabolol	NF-κB (IκBα blockade); PPARγ (activator)	Not quantified	Anti-inflammatory; anxiolytic in preclinical models	Kim et al. 2011; Jenkins et al. 2023
Terpinolene	No GPCR data published	—	Pharmacologically uncharacterized despite defining a distinct cannabis chemotype	—
β-Ocimene	No GPCR data published	—	Pharmacologically uncharacterized	—

II. Inhalation Pharmacokinetics Demonstrate Rapid CNS Access and Dose-Dependent Cognitive Effects

The therapeutic relevance of the receptor pharmacology described in Section I depends entirely on whether inhaled terpenes reach their molecular targets at sufficient concentrations to produce receptor engagement. This question—of bioavailability, distribution kinetics, and central nervous system access—is the pharmacokinetic bridge connecting in vitro receptor data to in vivo behavioral and clinical effects. The evidence reviewed here establishes three critical points: first, that inhaled monoterpenes achieve rapid systemic absorption with pulmonary uptake efficiencies of 60–65%, reaching peak brain concentrations within 30–90 minutes; second, that circulating terpene levels correlate directly with cognitive performance in healthy human volunteers, establishing dose-response relationships for inhaled terpene bioactivity; and third, that terpenes access the central nervous system through at least three distinct anatomical pathways, each with different pharmacokinetic profiles and functional consequences. Taken together, these data demonstrate that terpene inhalation—whether from essential oil diffusion, cannabis vaporization, or forest air exposure—is not aromatherapeutic theater but a pharmacologically meaningful route of drug delivery.

A. The Plasma–Cognition Correlation: Establishing Dose-Response for Inhaled Terpenes in Humans

The single most important study for the pharmacokinetic foundation of inhaled terpene bioactivity is Moss and Oliver (2012), which provided the first and, as of early 2026, still the only published demonstration that blood levels of an inhaled terpene correlate with cognitive performance in humans. Twenty healthy volunteers were exposed to rosemary essential oil aroma in a testing room while performing a battery of cognitive tasks; serum 1,8-cineole concentrations were subsequently measured by gas chromatography–mass spectrometry using terpinolene as an internal standard. The design was elegant in its simplicity: rather than

comparing aroma exposure versus control, it exploited the natural between-subject variation in terpene absorption to ask whether higher systemic concentrations predicted better performance within the exposed group.

The results established clear dose-response relationships. Pearson correlations between serum 1,8-cineole and cognitive outcomes revealed: serial subtraction of threes (accuracy) $r(18) = 0.469$, $p = 0.037$; serial subtraction of threes (reaction time) $r(18) = -0.502$, $p = 0.024$; and serial subtraction of sevens (accuracy) $r(18) = 0.433$, $p = 0.056$, approaching significance. Effect sizes ($r^2 = 0.19–0.25$) were medium-to-large by Cohen's conventions—explaining 19–25% of variance in cognitive performance from a single environmental terpene exposure. Notably, rapid visual information processing (RVIP) accuracy showed no correlation with serum levels, and contentedness showed a significant negative correlation, suggesting that the cognitive-enhancing and mood-modulating effects of 1,8-cineole are mediated through pharmacologically independent pathways. This dissociation between cognitive and affective outcomes is consistent with the multi-target receptor profile of 1,8-cineole (TRPM8 agonism, AChE inhibition) engaging functionally distinct neural circuits.

The Moss and Oliver finding has profound implications for the unified framework. If rosemary aroma produces measurable serum concentrations of 1,8-cineole that correlate with cognitive enhancement, then the pharmacological effects attributed to rosemary essential oil in the aromatherapy literature, to eucalyptol-containing cannabis strains in the cannabis literature, and to 1,8-cineole-rich forest air in the forest medicine literature all share a common pharmacokinetic basis. The molecule enters the blood through the same pulmonary and olfactory routes regardless of botanical source, reaches the same central targets, and produces quantitatively similar effects at comparable tissue concentrations. The disciplinary label attached to the exposure context is pharmacologically irrelevant.

B. Pharmacokinetic Parameters for Major Terpenes: A Compilation from Human Inhalation Studies

Human pharmacokinetic data for inhaled terpenes, while not as extensive as for pharmaceutical agents, are sufficient to construct a coherent absorption-distribution-metabolism-elimination profile for the major compounds. The foundational study is Jäger et al. (1996), who established 1,8-cineole inhalation pharmacokinetics through serial blood sampling: time to maximum concentration (Tmax) of approximately 18 minutes, a rapid distribution phase with a half-life ($t_{½α}$) of 6.7 minutes, and an elimination phase half-life ($t_{½β}$) of 104.6 minutes (approximately 1.7 hours). The elimination follows biphasic kinetics, and a clinically relevant sex difference emerged: female subjects showed approximately two-fold longer half-lives than males, attributed to the larger subcutaneous adipose compartment serving as a lipophilic reservoir. This sex difference in terpene pharmacokinetics has been largely overlooked in subsequent study designs but has potentially important implications for dose-response relationships in both therapeutic and recreational contexts.

For α-pinene, Falk and Hagberg (1990) conducted the definitive inhalation study in eight healthy males during controlled two-hour exposures at occupational-range concentrations. Pulmonary uptake was approximately 60%—meaning that nearly two-thirds of inhaled α-pinene crosses from alveolar air into the systemic circulation on first pass. Distribution was rapid ($t_{½α}$ approximately 5 minutes), consistent with α-pinene's high lipophilicity (logP approximately 4.8) facilitating rapid tissue partitioning. Primary elimination proceeded with $t_{½β}$ of 26–38 minutes, but a critical third phase was identified with $t_{½γ}$ of 695 minutes (approximately 11.6 hours), reflecting slow release from deeply sequestered adipose stores. This triphasic kinetics means that a single significant α-pinene inhalation exposure—whether from a walk through a pine forest, a rosemary essential oil diffusion session, or vaporization of an α-pinene-dominant cannabis strain—produces systemic terpene levels persisting for nearly 12 hours, well beyond the typical duration attributed to the acute exposure.

D-Limonene inhalation pharmacokinetics show similarly efficient pulmonary absorption at approximately 65% uptake across exposure concentrations from 10 to 450 mg/m³ (Kohlert et al., 2000). The general metabolic fate of inhaled monoterpenes follows a consistent pattern: rapid phase-I oxidation followed by phase-II glucuronide conjugation (primarily via CYP3A4 for 1,8-cineole) and renal elimination, with only 1–3% of the absorbed dose eliminated unchanged via exhalation. The implication for cross-botanical comparison is that regardless of whether α-pinene enters the respiratory tract from forest air, an essential oil diffuser, or cannabis vapor, the same metabolic pathway processes the same molecule, producing the same circulating metabolites at concentrations proportional to the inhaled dose.

Table 2 compiles the available human inhalation pharmacokinetic parameters for the five terpenes with published data. Several features merit emphasis. First, all major monoterpenes show pulmonary uptake efficiencies between 60–70%, far exceeding typical oral bioavailability for lipophilic compounds. Second, distribution half-lives of 5–7 minutes indicate near-immediate systemic delivery. Third, the presence of deep compartment phases ($t_{½γ}$ of 11.6 hours for α-pinene) means that repeated daily exposures—as might occur in an occupational, therapeutic, or recreational context—would produce accumulation in adipose tissue with correspondingly prolonged pharmacodynamic effects. Fourth, the sex-dependent kinetics documented for 1,8-cineole raise the possibility that female users may experience stronger or more prolonged terpene effects at equivalent exposures—a variable uncontrolled in most cannabis efficacy studies.

C. Blood-Brain Barrier Penetration: Physicochemical Prediction Meets Direct Measurement

The question of whether systemically absorbed terpenes reach the brain at pharmacologically relevant concentrations can be addressed through both physicochemical prediction and direct measurement. By the Lipinski-derived criteria for blood-brain barrier (BBB) penetration—molecular weight below 400 Da, lipophilicity (logP) greater than 1.5, polar surface area below

90 Å², and fewer than 8 hydrogen bonds—all major cannabis monoterpenes qualify comfortably. With molecular weights of 136–154 Da and logP values of 2.7–4.8, monoterpenes are among the most BBB-permeable natural products studied.

Direct measurement has confirmed these predictions with remarkable quantitative detail. Satou et al. (2016) measured terpene concentrations in mouse brain tissue following controlled inhalation exposures. α-Pinene reached maximal brain concentrations within 30 minutes of inhalation at levels exceeding 90 μg/g brain tissue—a concentration that substantially exceeds the AChE IC_{50} and BDNF-inducing thresholds described in Section I. Limonene reached peak brain levels exceeding 50 μg/g at 90 minutes, and linalool achieved similar concentrations on the same timescale. These are not trace detections; they represent pharmacologically significant brain tissue concentrations achieved through passive inhalation.

A finding from the Satou study with particular relevance for the cross-botanical framework: α-pinene brain concentrations were approximately two-fold higher when the compound was inhaled as a component of a mixed essential oil blend compared to single-compound inhalation at equivalent air concentrations. This synergistic absorption effect—possibly mediated by co-inhaled terpenes modulating pulmonary epithelial permeability, inhibiting first-pass metabolism, or altering protein binding—suggests that the complex terpene mixtures encountered in natural exposure scenarios (essential oil blends, whole cannabis vapor, forest air) may produce higher brain concentrations than would be predicted from single-compound pharmacokinetic data. If confirmed in human studies, this finding would have direct implications for formulation design and would provide a pharmacokinetic rationale for the superiority of whole-plant cannabis preparations over single-compound isolates.

Additional evidence for CNS access comes from Pries et al. (2023), who documented a BBB-opening effect of 1,8-cineole—suggesting that cineole-containing essential oils or cannabis strains may transiently increase paracellular permeability, potentially enhancing the brain delivery of co-administered compounds. MacKenzie et al. (2023) provided the first demonstration of systemic distribution of orally administered 1,8-cineole to peripheral target

tissues (nasal polyp tissue in chronic rhinosinusitis patients), confirming that even the oral route delivers cineole to sites distant from the gastrointestinal tract. Together, these studies establish that the question is not whether inhaled terpenes reach the brain, but at what concentrations, through which routes, and with what temporal profiles—the answers to which are now quantitatively defined for the major compounds.

D. Three Nose-to-Brain Pathways: Anatomical Substrates for Terpene CNS Effects

The nasal delivery of volatile compounds to the central nervous system proceeds through three anatomically and pharmacokinetically distinct pathways, each with different implications for terpene bioactivity. Understanding these routes is essential for interpreting the behavioral and clinical effects documented in both the aromatherapy and cannabis inhalation literatures, and for designing the rational terpene delivery strategies that a unified framework makes possible.

The first pathway—olfactory nerve transport—begins at the olfactory epithelium in the superior nasal cavity. Volatile terpenes, by virtue of their lipophilicity and small molecular size, can traverse the olfactory epithelium either transcellularly (through sustentacular support cells) or perineurally (along the olfactory nerve sheaths) to reach the olfactory bulb. From there, direct neural projections deliver compounds to the piriform cortex, amygdala, hippocampus, and entorhinal cortex—structures collectively constituting the limbic system. This pathway is rapid (onset within minutes) and bypasses both the blood-brain barrier and first-pass hepatic metabolism. The Harada et al. (2018) anosmia experiment (Section I.C) demonstrated that linalool's anxiolytic effects depend entirely on this olfactory route, establishing that at least for some terpene-mediated behaviors, olfactory nerve transport is not merely a delivery route but the mechanistic pathway through which effects are produced.

The second pathway—trigeminal nerve transport—involves terpene absorption through the respiratory and olfactory epithelium followed by transport along branches of the trigeminal nerve (V1 ophthalmic, V2 maxillary) to the brainstem trigeminal nucleus. This route is

particularly significant because it remains intact in many anosmic patients (Huang et al., 2023; Qiu, 2025), meaning that individuals who have lost their sense of smell may still experience central terpene effects through trigeminal delivery. The trigeminal pathway delivers compounds to brainstem structures involved in pain modulation, autonomic regulation, and arousal—functionally distinct from the limbic structures accessed by the olfactory route. Several terpenes, including 1,8-cineole and menthol, are known trigeminal stimulants, engaging TRPM8 and TRPA1 channels on trigeminal nerve endings. The contribution of trigeminal transport to the overall CNS terpene delivery in human inhalation scenarios has not been quantified, representing a significant gap in the pharmacokinetic understanding of essential oil and cannabis aromatherapy.

The third pathway—systemic respiratory absorption—is the classical pharmacokinetic route: terpenes absorb through the highly vascularized respiratory epithelium of the nasal cavity and lower airways into the pulmonary circulation, enter the systemic blood, and cross the BBB to reach central targets. This is the pathway characterized by the Falk and Hagberg, Jäger, and Kohlert pharmacokinetic studies described above, with pulmonary uptake efficiencies of 60–65% and Tmax values of 18–30 minutes. Unlike the olfactory pathway, systemic absorption is subject to first-pass pulmonary metabolism and plasma protein binding, and delivers terpenes to the brain through a generalized distribution pattern rather than targeting specific limbic structures.

For volatile monoterpenes at 136–154 Da, all three pathways operate simultaneously during any inhalation exposure. Paracellular transport through tight junctions (optimal for compounds below 1 kDa) and direct transcellular passage through lipophilic membranes are both available at the nasal epithelium. The relative contribution of each pathway depends on anatomical variables (nasal airflow patterns, epithelial integrity, mucus layer thickness), physicochemical properties of the individual terpene (lipophilicity, vapor pressure, aqueous solubility), and exposure parameters (concentration, duration, breathing pattern). No study has simultaneously quantified all three pathways for any terpene in humans. However, the functional

consequence of this multi-route delivery system is that inhaled terpenes access the CNS through both direct neural pathways (producing rapid, limbic-targeted effects) and systemic absorption (producing delayed, globally distributed effects)—a dual-access pharmacokinetic profile with no analog in conventional oral pharmacotherapy.

E. The Moss Research Program (2003–2025): Two Decades of Inhaled Terpene Cognitive Pharmacology

The most systematic investigation of inhaled terpene effects on human cognition is the research program conducted by Mark Moss and colleagues at Northumbria University, spanning more than two decades and encompassing over 400 participants across multiple randomized controlled studies. This body of work, published in neuroscience and psychopharmacology journals, is essentially unknown to the cannabis pharmacology community—a disciplinary blind spot that exemplifies the fragmentation this review addresses.

The program began with Moss and Cook (2003), who randomized 144 healthy adults to rosemary aroma exposure, lavender aroma exposure, or an unscented control condition during cognitive testing. Rosemary enhanced Quality of Memory and Secondary (long-term) Memory ($r^2 = 0.06–0.07$, small effect sizes), while lavender—consistent with its linalool-mediated GABA-A modulation—impaired Working Memory. This dissociation between terpene profiles and cognitive domains was the first indication that different terpene compositions produce qualitatively different cognitive signatures rather than a unitary "aromatherapy effect."

The 2012 plasma-correlation study (Section II.A) elevated this work from behavioral observation to mechanistic pharmacology by demonstrating that the cognitive effects tracked with circulating terpene concentrations, establishing dose-response relationships with medium-to-large effect sizes. Moss et al. (2017) added prospective memory—the ability to remember to perform intended actions in the future—to the list of cognitive domains enhanced by rosemary aroma, with effect sizes comparable to those seen for retrospective memory.

The culmination of this program is Moss et al. (2025), a double-blind, positive-controlled trial in 90 participants that introduced functional near-infrared spectroscopy (fNIRS) brain imaging. Participants were randomized to a complex essential oil blend (branded "Genius," containing patchouli, neroli, grapefruit, cardamom, frankincense, spikenard, rosemary, and lemongrass), sage essential oil, or a control condition. Both the Genius blend and sage enhanced memory and executive function, with the Genius blend showing larger effects. The critical advance: fNIRS revealed enhanced prefrontal cortex oxygenated hemoglobin during cognitive tasks in both treatment conditions. This represents the first simultaneous demonstration that essential oil inhalation enhances both cognitive performance and brain metabolic activity, providing a hemodynamic correlate to the behavioral effects documented over the preceding two decades.

The Moss program establishes several principles essential for the unified framework. First, inhaled terpenes produce reliable, replicable cognitive effects in healthy humans at environmental exposure concentrations—not pharmacological doses. Second, different terpene profiles produce qualitatively different cognitive signatures, consistent with their distinct receptor pharmacologies. Third, these effects scale with blood concentrations, confirming pharmacokinetic mediation rather than purely psychological expectancy. Fourth, multi-terpene blends can produce larger effects than single essential oils, consistent with the network pharmacology principle that multi-target engagement outperforms single-target intervention. Cannabis vaporization delivers the same terpene compounds at substantially higher concentrations than ambient aroma exposure, yet the cannabis pharmacology literature has not systematically investigated terpene-specific cognitive effects independent of cannabinoid activity—a translational gap that the unified framework highlights.

F. Concentrated Delivery Systems: Evidence from the AromaStick Research Program

The exposure concentration achievable in terpene inhalation varies enormously across delivery modalities—from micrograms per cubic meter in forest air to milligrams per cubic meter in essential oil diffusion to potentially much higher concentrations in direct inhalation devices. Schneider et al. (2019) reported results across 465 total participants using an AromaStick® inhaler that delivers concentrated essential oils directly to one nostril, representing a delivery modality intermediate between ambient diffusion and cannabis vaporization in terms of delivered dose.

The physiological effects documented across this participant pool were broad and rapid: blood pressure changes, heart rate modulation (with the inhaler up to three-fold more effective than room spray delivery at the same essential oil composition), arterial oxygen saturation increased 2.5-fold above normal deep breathing alone, salivary cortisol decreased, urinary serotonin increased, pain onset relief accelerated with duration and intensity reduced by up to 30%, and visual scanning speed enhanced by up to 29%. The investigators described effects as occurring almost immediately and being clinically relevant in magnitude. An important caveat applies: these studies were funded by the AromaStick manufacturer, and the absence of independent replication warrants caution in interpreting the absolute magnitude of effects. Nevertheless, the dose-dependency across delivery methods (room spray versus direct inhaler) provides pharmacokinetic plausibility, and the direction of effects is consistent with the receptor pharmacology and cognitive studies reviewed above.

The relevance for the unified framework lies in the concentration gradient across exposure scenarios. Forest air delivers terpenes at 3.5–56.0 μg/m³ total concentration. Ambient essential oil diffusion typically produces room-air concentrations one to two orders of magnitude higher. Direct nasal inhalation devices increase delivered concentrations by another order of magnitude. Cannabis vaporization, which rapidly heats terpene-containing plant material or

concentrates, likely delivers the highest per-puff terpene doses of any common exposure scenario, though systematic measurement of terpene delivery from cannabis vaporization devices is surprisingly sparse. The biological effects documented across this concentration spectrum—from Li's forest immunology at the lowest concentrations through Moss's cognitive enhancement at intermediate concentrations to the Schneider physiological data at higher concentrations—collectively describe a dose-response continuum for inhaled terpene bioactivity that spans five orders of magnitude of delivered concentration while engaging the same molecular targets.

G. Section Summary: Pharmacokinetic Validation of the Inhalation Route

The pharmacokinetic evidence reviewed in this section establishes that terpene inhalation is a pharmacologically valid delivery route with quantified absorption, distribution, metabolism, and elimination parameters for the major compounds. Pulmonary uptake efficiencies of 60–65%, peak brain concentrations within 30–90 minutes, direct correlation between blood levels and cognitive performance, multi-pathway CNS access, and dose-response relationships across five orders of magnitude of exposure concentration collectively validate the biological plausibility of inhaled terpene pharmacology.

Two critical gaps remain. First, no human study has simultaneously measured terpene concentrations at the olfactory epithelium, in the cerebrospinal fluid, and in the systemic circulation during inhalation exposure—meaning that the relative contributions of the olfactory, trigeminal, and systemic pathways to central terpene effects remain unquantified. Second, the pharmacokinetics of terpene inhalation from cannabis vaporization specifically have not been characterized: the temperature profiles, co-delivered cannabinoids, and carrier matrix of cannabis vapor are distinct from essential oil diffusion, and the assumption that equivalent pharmacokinetic parameters apply may not hold. Addressing these gaps is essential for translating the unified pharmacological framework into evidence-based dosing recommendations for terpene-enhanced cannabis therapy.

Table 2. Human inhalation pharmacokinetic parameters for major terpenes. Tmax, time to maximum concentration; t½α, distribution half-life; t½β, elimination half-life; t½γ, deep-compartment half-life. Brain concentrations from mouse direct-measurement studies (Satou et al. 2016). Female subjects show ~2× longer half-lives for 1,8-cineole (Jäger et al. 1996). α-Pinene brain concentrations were 2-fold higher from mixed essential oil blends vs. single-compound inhalation.

---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- Terpene Pulmonary Uptake Tmax t½α t½β t½γ Peak Brain Conc. Key Reference(s)

α-Pinene ~60% ~30 min (brain) ~5 min 26–38 min 695 min (~11.6 h) >90 µg/g (mouse, 30 min) Falk & Hagberg 1990; Satou et al. 2016

1,8-Cineole Not reported ~18 min (serum) 6.7 min 104.6 min (~1.7 h) Not reported Not directly measured Jäger et al. 1996; Moss & Oliver 2012

d-Limonene ~65% ~90 min (brain) Not reported Not reported Not reported >50 µg/g (mouse, 90 min) Kohlert et al. 2000; Satou et al. 2016

Linalool Not reported ~90 min (brain) Not reported Not reported Not reported Similar to limonene (mouse) Satou et al. 2016 ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------

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Figure 2. Three nose-to-brain pathways for inhaled terpene CNS delivery. Pathway 1 (olfactory nerve): transcellular/perineural transport through the cribriform plate to the olfactory bulb and limbic structures; fastest onset, bypasses BBB; linalool anxiolysis depends entirely on this route (Harada et al., 2018). Pathway 2 (trigeminal nerve): absorption through nasal epithelium and transport via V1/V2 branches to brainstem; intact in anosmic patients; engages pain modulation and autonomic centers. Pathway 3 (systemic respiratory): classical

pharmacokinetic route via pulmonary circulation, BBB crossing, and generalized CNS distribution; characterized by 60–65% pulmonary uptake, Tmax 18–90 min, and up to 11.6 h deep-compartment residence. Right panel: exposure concentration gradient spanning five orders of magnitude across delivery modalities, from forest air (3.5–56.0 μg/m³) to cannabis vaporization, all engaging the same molecular targets through the same anatomical pathways.

Section 2 word count: ~3,900 words | Cumulative (Sections 1–2): ~7,600 words of target 15,000–25,000

Table 2. Human inhalation pharmacokinetic parameters for major terpenes. Tmax, time to maximum concentration; t½α, distribution half-life; t½β, elimination half-life; t½γ, deep-compartment half-life. Brain concentrations from mouse direct-measurement studies (Satou et al. 2016). Female subjects show ~2× longer half-lives for 1,8-cineole (Jäger et al. 1996). α-Pinene brain concentrations were 2-fold higher from mixed essential oil blends vs. single-compound inhalation.

Terpene	**Pulmonary Uptake**	**Tmax**	**t½α**	**t½β**	**t½γ**	**Peak Brain Conc.**	**Key Reference(s)**
α-Pinene	~60%	~30 min (brain)	~5 min	26–38 min	695 min (~11.6 h)	>90 μg/g (mouse, 30 min)	Falk & Hagberg 1990; Satou et al. 2016
1,8-Cineole	Not reported	~18 min (serum)	6.7 min	104.6 min (~1.7 h)	Not reported	Not directly measured	Jäger et al. 1996; Moss & Oliver 2012
d-Limonene	~65%	~90 min (brain)	Not reported	Not reported	Not reported	>50 μg/g (mouse, 90 min)	Kohlert et al. 2000; Satou et al. 2016
Linalool	Not reported	~90 min (brain)	Not reported	Not reported	Not reported	Similar to limonene (mouse)	Satou et al. 2016

All three pathways operate simultaneously during any inhalation exposure. Relative contribution depends on:
anatomical variables (nasal airflow, epithelial integrity) · physicochemical properties (logP, vapor pressure) · exposure parameters (concentration, duration, breathing pattern)
No human study has simultaneously quantified all three pathways for any terpene. (Synthesized from Huang et al. 2023; Qiu 2025; Falk & Hagberg 1990; Harada et al. 2018)

III. Aromatherapy Randomized Controlled Trials Organized by Terpene Reveal a Substantial but Quality-Limited Evidence Base

The aromatherapy clinical literature comprises thousands of studies published over five decades, yet its value for pharmacological analysis has been limited by a persistent organizational flaw: trials are categorized by essential oil (lavender, rosemary, bergamot) rather than by active molecular constituent. This taxonomic convention—naming the botanical source rather than the pharmacological agent—would be unacceptable in any other domain of clinical pharmacology. No reviewer of analgesic trials would organize data by "willow bark studies" versus "meadowsweet studies" rather than by their shared active compound, salicylate. Yet the aromatherapy literature has perpetuated precisely this framework, obscuring the convergent pharmacology of shared terpene constituents across different botanical preparations.

This section reorganizes the clinical trial evidence by primary terpene constituent, revealing patterns invisible in the traditional essential oil–based taxonomy. Three terpene systems dominate the clinical evidence: linalool (studied primarily through lavender essential oil), 1,8-cineole (studied through rosemary and eucalyptus oils), and limonene (studied through citrus oils). For each, we review the scope and quality of randomized controlled trial evidence, identify the strongest individual studies, compile aggregate participant numbers and effect sizes where meta-analyses exist, and assess the critical methodological limitations that constrain interpretation. The Silexan oral lavender preparation receives extended treatment as the only terpene-based formulation to have completed a rigorous Phase III clinical trial program meeting conventional pharmaceutical standards.

A. Linalool and Lavender Inhalation: 972 Participants Across 11 Recent RCTs, With Systematic Quality Concerns

Lavender essential oil—containing linalool (25–46%) and linalyl acetate (25–57%) as its dominant constituents—is by far the most extensively studied essential oil in clinical research.

The most recent systematic review specifically addressing inhalation delivery is Alhazmi et al. (2023), which identified 11 randomized controlled trials published between 2018 and 2022 evaluating lavender essential oil inhalation for anxiety across diverse clinical populations. These 11 trials enrolled a combined 972 participants, of whom 431 received lavender inhalation. Ten of the 11 trials reported statistically significant reductions in anxiety measures. The clinical populations were heterogeneous, spanning pre-surgical anxiety, hemodialysis-associated distress, electroconvulsive therapy apprehension, myofascial pain, intrauterine insemination procedures, postmenopausal depression, pediatric dental anxiety, and anxiety in older adults. This breadth of positive findings across widely varying contexts is consistent with a genuine anxiolytic pharmacological effect rather than a context-specific placebo response—though the absence of a pooled meta-analytic effect size from this review limits quantitative comparison.

Quantitative meta-analytic data are available from two larger reviews that included both inhalation and other delivery routes. Kang et al. (2019) meta-analyzed 22 RCTs of lavender aromatherapy and found clinically meaningful effect sizes across multiple physiological and psychological endpoints: anxiety reduction with a Hedges' g of −0.65 (95% CI: −0.84 to −0.46), representing a moderate-to-large effect; heart rate reduction g = −0.53; systolic blood pressure reduction g = −0.22; salivary cortisol reduction g = −1.29, a large effect; and salivary chromogranin A reduction g = −2.29, a very large effect. The gradient from moderate psychological effects to very large biochemical effects suggests that the physiological response to lavender inhalation is more robust and consistent than the subjective psychological experience—consistent with a genuine pharmacological mechanism operating through autonomic and neuroendocrine pathways rather than purely cognitive-evaluative processes.

The largest systematic effort is Donelli et al. (2019), encompassing 102 RCTs and 25 non-randomized studies involving more than 13,000 subjects across all lavender delivery routes. Quantitative meta-analysis of oral lavender (80 mg/day for six weeks or longer) found a Hamilton Anxiety Scale reduction of MD −2.90 (95% CI: −4.86 to −0.95, p = 0.004). For inhalation specifically, the pooled effect size was Hedges' g = −0.73 (95% CI: −1.00 to −0.46, p

< 0.00001)—a medium-to-large effect consistent with the Kang estimate. However, the Donelli review included a critical cautionary assessment that is frequently omitted when these data are cited: the authors characterized the evidence as having low average quality and high overall risk of bias, concluding that it would be premature to consider lavender essential oil inhalation as an established treatment for anxiety based on the available evidence.

The quality concerns are not trivial. Blinding in inhalation aromatherapy trials is inherently difficult: lavender has a distinctive aroma recognizable to most participants, making true double-blinding nearly impossible without elaborate controls (such as masking with other scents or using encapsulated delivery). Expectancy effects are substantial in aromatherapy contexts, where cultural associations between lavender and relaxation are deeply embedded. Sample sizes in individual trials are typically small (median approximately 40–60 participants), attrition rates are inconsistently reported, and outcome measures vary widely across studies—making cross-trial comparison and meta-analytic pooling methodologically challenging. These are not disqualifying limitations; they are characteristic of a field in an early evidence-development phase. But they demand that the effect sizes be interpreted as preliminary estimates of therapeutic potential rather than definitive measures of pharmacological efficacy.

B. Silexan: The Gold-Standard Anxiolytic Terpene Formulation and Its Implications for Cannabis Terpene Research

The Silexan clinical program stands apart from the broader aromatherapy literature in methodological rigor, representing the only terpene-based formulation to have completed a conventional pharmaceutical development trajectory including Phase III double-blind, placebo-controlled, active-comparator trials. Silexan (marketed as Lasea® in Germany) is a standardized oral preparation of Lavandula angustifolia essential oil, delivered in enteric-coated soft gelatin capsules at 80 mg or 160 mg daily. Its terpene composition—predominantly linalool and linalyl acetate—is qualitatively identical to the compounds studied in inhalation aromatherapy, but the

oral delivery route, standardized dosing, and pharmaceutical-grade trial design eliminate many of the methodological limitations that plague the inhalation literature.

The pivotal trial is Kasper et al. (2014), a study whose results deserve careful examination because they challenge conventional assumptions about the pharmacological potency of plant terpenes. In this multicenter trial, 539 adults meeting DSM-IV-TR criteria for generalized anxiety disorder were randomized in a 1:1:1:1 ratio to Silexan 160 mg/day (n = 121), Silexan 80 mg/day (n = 135), paroxetine 20 mg/day (n = 132), or placebo (n = 135) for 10 weeks. The Hamilton Anxiety Scale reductions were: Silexan 160 mg, −14.1 ± 9.3 points ($p < 0.01$ versus placebo); Silexan 80 mg, −12.8 ± 8.7 points ($p < 0.01$ versus placebo); paroxetine 20 mg, −11.3 ± 8.0 points ($p = 0.10$ versus placebo in the full analysis set). Responder rates, defined as 50% or greater HAMA reduction, were 60.3% for Silexan 160 mg, 51.9% for Silexan 80 mg, 43.2% for paroxetine, and 37.8% for placebo. Remission rates (HAMA below 10) were 46.3% for high-dose Silexan versus 34.1% for paroxetine.

Two aspects of these results demand emphasis. First, a standardized preparation of plant terpenes numerically outperformed a first-line SSRI antidepressant on every measured endpoint in a properly powered, double-blind trial—a finding that should command attention from any pharmacologist, regardless of disciplinary orientation. Second, Silexan produced no withdrawal symptoms upon discontinuation, a clinically significant advantage over SSRIs where discontinuation syndrome affects a substantial proportion of patients. The adverse event incidence density was comparable to or lower than placebo, with gastrointestinal complaints (primarily eructation with lavender taste) as the most common treatment-emergent effect.

The aggregate evidence from the complete Silexan program was meta-analyzed by Dold et al. (2023), pooling all five completed double-blind RCTs (total n = 1,213, Silexan 80 mg/day versus placebo, 10 weeks). The responder rate ratio was 1.34 favoring Silexan; the proportion rated "much improved" or "very much improved" on the Clinical Global Impressions scale showed a rate ratio of 1.51. Superiority was observed across all diagnostic categories: subthreshold anxiety, mixed anxiety-depressive disorder, and generalized anxiety disorder. The

earlier Woelk and Schläfke (2010) trial had demonstrated that Silexan 80 mg was comparable to lorazepam 0.5 mg over six weeks—establishing equivalence with a benzodiazepine as well as superiority over an SSRI. On the strength of this evidence, Silexan was included in the Bandelow et al. (2023) clinical practice guidelines as a recommended medication for anxiety disorders and is authorized for prescription in Germany.

The implications of the Silexan program for the unified framework are profound. The same linalool molecule that cannabis researchers study as a minor terpene in lavender-scented strains, that forest medicine researchers encounter as a component of forest air, and that aromatherapists apply through inhalation has been demonstrated, at pharmaceutical-grade evidence standards, to outperform a first-line psychiatric medication for anxiety. This is not an extrapolation from rodent models or in vitro receptor binding; it is a direct clinical demonstration in over 1,200 patients. Yet the Silexan data appear in psychiatric and phytotherapy journals and are essentially absent from the cannabis pharmacology literature. A PubMed search for "Silexan" retrieves no results in Cannabis and Cannabinoid Research, the Journal of Cannabis Research, or any cannabis-specific journal. This is perhaps the most striking example of the disciplinary fragmentation that the present review seeks to address: a pharmaceutical-grade demonstration of terpene anxiolytic efficacy, invisible to the field most actively pursuing terpene-cannabinoid interactions.

C. 1,8-Cineole and Rosemary: Cognitive Enhancement Across Multiple Domains

The clinical evidence for 1,8-cineole cognitive enhancement derives primarily from the Moss research program detailed in Section II.E, supplemented by respiratory medicine data demonstrating anti-inflammatory efficacy at therapeutic doses. The Moss series establishes enhancement across four cognitive domains: Quality of Memory and Secondary (long-term) Memory (Moss and Cook, 2003; effect sizes $r^2 = 0.06–0.07$); working memory speed and accuracy correlated with blood 1,8-cineole concentrations (Moss and Oliver, 2012; $r^2 = 0.19–$

0.25); prospective memory (Moss et al., 2017); and prefrontal cortex metabolic enhancement measured by functional near-infrared spectroscopy (Moss et al., 2025). The progression from small behavioral effects in 2003 to medium-large pharmacokinetically validated effects in 2012 to neuroimaging-confirmed metabolic changes in 2025 represents a maturing evidence base that has advanced from phenomenological observation to mechanistically grounded human pharmacology.

Complementary evidence from respiratory medicine reinforces the pharmacological credibility of 1,8-cineole. Juergens et al. (2003) demonstrated that oral 1,8-cineole at 600 mg/day achieves clinically meaningful anti-inflammatory effects in a double-blind, placebo-controlled trial of asthma patients: the cineole group achieved a 36% reduction in oral corticosteroid dose versus 7% in the placebo group ($p = 0.006$). While this respiratory application operates through different effector pathways (TRPM8-mediated airway anti-inflammation rather than AChE-mediated cognitive enhancement), it establishes that oral terpene delivery achieves systemically relevant concentrations with clinically measurable outcomes—bridging the gap between the inhalation cognitive studies and the receptor pharmacology detailed in Section I.

A notable limitation of the 1,8-cineole cognitive literature is its almost exclusive reliance on rosemary essential oil as the delivery vehicle, which contains 1,8-cineole alongside α-pinene, camphor, and other potentially neuroactive compounds. The Moss and Oliver correlation study partially addresses this confound by measuring serum 1,8-cineole specifically rather than relying on categorical rosemary-versus-control comparisons, but the contribution of co-inhaled terpenes to the observed cognitive effects cannot be excluded. This limitation is structurally analogous to the challenge facing cannabis terpene research, where individual terpene effects must be disentangled from the complex chemotype matrix—and the analytical approach used by Moss (measuring individual compound blood levels rather than categorizing by source material) provides a methodological template directly applicable to cannabis terpene pharmacology.

D. Limonene and Citrus Oils: The Hedonic Valence Confound

The clinical evidence for limonene's mood-modulating effects through citrus essential oil inhalation is compromised by a fundamental confound that has only recently been clearly articulated. The positive evidence includes Watanabe et al. (2015), who found in a randomized crossover study of 41 healthy females that bergamot essential oil inhalation (containing approximately 25–53% limonene) significantly reduced salivary cortisol, increased parasympathetic high-frequency heart rate variability ($p = 0.003$), and improved negative mood ratings. Komori et al. (1995) reported that citrus fragrance vaporization in 12 depressive patients reduced antidepressant medication requirements and normalized natural killer cell activity and cortisol levels.

However, the Komori study—despite being one of the most frequently cited papers in the aromatherapy-mood literature—is methodologically weak: it lacked a control group, employed no blinding, and enrolled only 12 participants. More fundamentally, Hoenen et al. (2016) conducted a properly controlled study in 78 participants and found that d-limonene did not directly elevate mood compared to controls. Instead, mood improvement was associated with positive subjective judgment of the odor—suggesting that hedonic valence (whether participants liked the smell) rather than direct pharmacological action may mediate the mood effects attributed to citrus aromatherapy.

This hedonic valence confound represents a critical methodological challenge for the entire aromatherapy field. If mood effects are mediated by the pleasantness of the aroma rather than by specific receptor activation, then the clinical relevance is not diminished (pleasant environmental stimuli improving mood is a legitimate therapeutic mechanism), but the pharmacological interpretation changes fundamentally. The resolution may lie in the A2a receptor pharmacology: Park et al.'s (2011, 2021) demonstrations of limonene's A2a agonism and A2a-dependent anxiolysis in animal models suggest genuine receptor-mediated effects at pharmacological concentrations, while the Hoenen finding suggests that ambient inhalation

concentrations may be insufficient to engage these receptors, leaving hedonic valence as the dominant mechanism at low doses. This dose-dependent mechanistic transition—from hedonic at low concentrations to pharmacological at higher concentrations—would predict that limonene delivered through cannabis vaporization (at far higher per-puff doses than ambient citrus diffusion) might produce direct A2a-mediated effects that ambient citrus aromatherapy does not. The Spindle et al. (2024) finding of selective anxiolysis when limonene is co-vaporized with THC is consistent with this prediction.

E. Aggregate Evidence Assessment: Estimating the Total Clinical Evidence Base

A comprehensive accounting of the aromatherapy inhalation clinical trial literature reveals a larger evidence base than is typically appreciated. Based on the Donelli et al. (2019) aggregate of more than 13,000 participants across all lavender delivery routes, the O'Hara (2023) identification of 4,681 participants across 50 inhalation RCTs specifically addressing stress and anxiety, and additional trials in pain, insomnia, cognition, and nausea not captured in anxiety-focused reviews, the total across all published aromatherapy inhalation RCTs likely falls in the range of 15,000 to 25,000 participants. This is not a trivial evidence base; it exceeds the total clinical trial population for many approved pharmaceutical agents. The Silexan program alone contributes 1,213 participants in high-quality double-blind RCTs—a number comparable to the registration trial populations for several anxiolytic medications.

Nevertheless, the quality distribution is heavily skewed. The Silexan trials meet conventional pharmaceutical standards; the Moss cognition studies employ rigorous designs with pharmacokinetic validation; a modest number of other trials use adequate randomization, blinding, and sample sizes. But the bulk of the literature consists of small ($n < 60$), inadequately blinded studies with high risk of bias, inconsistent outcome measures, and limited follow-up. The field has generated volume without proportionate quality. For the unified framework, this means that the clinical evidence for inhaled terpene bioactivity should be considered established in

principle (the Silexan and Moss data alone are sufficient for this) but inadequately characterized in detail for most specific terpene-endpoint combinations. The aromatherapy RCT literature is best interpreted not as a definitive clinical evidence base but as a large-scale preliminary signal indicating which terpene-endpoint combinations merit rigorous investigation—including investigation in the context of cannabis co-administration, where the entourage effect hypothesis predicts enhanced efficacy.

F. Section Summary: What the Aromatherapy Literature Tells Cannabis Pharmacology

The aromatherapy clinical trial evidence, reorganized by terpene constituent rather than essential oil source, yields several conclusions directly relevant to cannabis pharmacology. First, linalool is a clinically validated anxiolytic in human patients, with effect sizes meeting or exceeding those of first-line psychiatric medications in the Silexan program. Any cannabis strain containing significant linalool concentrations is delivering a compound with established anxiolytic efficacy to the consumer—a pharmacological reality unacknowledged in dispensary labeling or clinical cannabis guidance. Second, 1,8-cineole produces dose-dependent cognitive enhancement in humans, with effects tracking blood concentrations—meaning that cannabis strains with cineole content are delivering a cognitive-active compound whose effects can be quantified through serum measurement. Third, limonene's mood effects at ambient inhalation concentrations may be predominantly hedonic rather than pharmacological, but its pharmacological effects at higher concentrations (A2a-mediated anxiolysis, selective attenuation of THC-induced anxiety) suggest a dose-dependent transition from aroma-mediated to receptor-mediated mechanisms that cannabis delivery may cross.

The methodological template from the best aromatherapy studies—particularly the Moss plasma-correlation approach and the Silexan pharmaceutical-grade trial design—provides directly transferable methods for cannabis terpene research. Conversely, the quality limitations

of the broader aromatherapy literature provide cautionary lessons: cannabis terpene studies must prioritize adequate blinding (using terpene-matched placebos), pharmacokinetic validation (measuring circulating terpene levels rather than relying on categorical exposure conditions), and sufficient statistical power (sample sizes calculated from the medium effect sizes now established in the Moss and Silexan data). The 15,000–25,000 participants enrolled in aromatherapy inhalation trials represent an underutilized pharmacological resource: they have been inhaling the same molecules that cannabis consumers inhale, with clinical outcomes measured and published. A unified analysis of this evidence alongside the emerging cannabis terpene trial data would substantially accelerate the characterization of terpene bioactivity in humans.

Table 3. Aromatherapy randomized controlled trial evidence organized by primary terpene constituent. DB-RCT, double-blind randomized controlled trial; HAMA, Hamilton Anxiety Scale; RR, rate ratio; CGI, Clinical Global Impressions; HF-HRV, high-frequency heart rate variability; PFC, prefrontal cortex; fNIRS, functional near-infrared spectroscopy. Aggregate participant estimate based on Donelli 2019 (>13,000 all lavender routes), O'Hara 2023 (4,681 across 50 inhalation RCTs), plus additional domains.

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(Donelli); blinding inherently difficult Alhazmi et al. 2023; Kang et al. 2019; Donelli et al. 2019

Linalool (Silexan oral) 80–160 mg/day enteric capsules 1,213 (5 DB-RCTs) HAMA −14.1 pts (160 mg); responder RR = 1.34; CGI RR = 1.51; comparable to lorazepam 0.5 mg Pharmaceutical-grade DB-RCT; active comparator (paroxetine); included in German clinical guidelines Kasper et al. 2014; Dold et al. 2023; Woelk & Schläfke 2010

1,8-Cineole (rosemary) Ambient aroma; oral (600 mg/day for respiratory) ~400+ (Moss series); asthma RCT (Juergens) r^2 = 0.19–0.25 cognition (Moss 2012); 36% steroid sparing (Juergens); fNIRS PFC enhancement (Moss 2025) Dose-response validated by plasma levels; small–medium-large effect sizes; single research group for cognition Moss & Oliver 2012; Moss et al. 2025; Juergens et al. 2003

d-Limonene (citrus oils) Ambient aroma (bergamot, citrus) ~130 (key studies) Cortisol reduction and HF-HRV increase (Watanabe); no direct mood elevation vs. control (Hoenen) Hedonic valence confound (Hoenen 2016); Komori (n=12) uncontrolled; dose-dependent mechanism transition likely Watanabe et al. 2015; Hoenen et al. 2016; Komori et al. 1995

Aggregate (all terpenes, all inhalation RCTs) Inhalation (all modalities) Est. 15,000–25,000 Variable; strongest for linalool anxiety and 1,8-cineole cognition Quality heavily skewed: Silexan/Moss rigorous; bulk of literature small, poorly blinded, high bias risk Donelli et al. 2019; O'Hara 2023

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Section 3 word count: ~3,400 words | Cumulative (Sections 1–3): ~10,300 words of target 15,000–25,000

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Primary Terpene	**Delivery / Source**	**N Participants**	**Key Effect Sizes**	**Quality Assessment**	**Key Reference(s)**
Linalool (lavender inhalation)	Ambient aroma / diffuser	972 (11 RCTs); >13,000 (all routes)	g = −0.65 anxiety (Kang); g = −0.73 inhalation (Donelli); cortisol g = −1.29	10/11 trials positive; low average quality; high risk of bias (Donelli); blinding inherently difficult	Alhazmi et al. 2023; Kang et al. 2019; Donelli et al. 2019
Linalool (Silexan oral)	80–160 mg/day enteric capsules	1,213 (5 DB-RCTs)	HAMA −14.1 pts (160 mg); responder RR = 1.34; CGI RR = 1.51; comparable to lorazepam 0.5 mg	Pharmaceutical-grade DB-RCT; active comparator (paroxetine); included in German clinical guidelines	Kasper et al. 2014; Dold et al. 2023; Woelk & Schläfke 2010

1,8-Cineole (rosemary)	Ambient aroma; oral (600 mg/day for respiratory)	~400+ (Moss series); asthma RCT (Juergens)	$r^2 = 0.19–0.25$ cognition (Moss 2012); 36% steroid sparing (Juergens); fNIRS PFC enhancement (Moss 2025)	Dose-response validated by plasma levels; small–medium-large effect sizes; single research group for cognition	Moss & Oliver 2012; Moss et al. 2025; Juergens et al. 2003
d-Limonene (citrus oils)	Ambient aroma (bergamot, citrus)	~130 (key studies)	Cortisol reduction and HF-HRV increase (Watanabe); no direct mood elevation vs. control (Hoenen)	Hedonic valence confound (Hoenen 2016); Komori (n=12) uncontrolled; dose-dependent mechanism transition likely	Watanabe et al. 2015; Hoenen et al. 2016; Komori et al. 1995
Aggregate (all terpenes, all inhalation RCTs)	Inhalation (all modalities)	Est. 15,000–25,000	Variable; strongest for linalool anxiety and 1,8-cineole cognition	Quality heavily skewed: Silexan/Moss rigorous; bulk of literature small, poorly blinded, high bias risk	Donelli et al. 2019; O'Hara 2023

Section 4: The Entourage Effect

Pharmacological Reviews (ASPET)

Prepared for submission to Pharmacological Reviews (ASPET)

IV. The Entourage Effect: From Endocannabinoid Biochemistry to the A2a Mechanism

A. Origin and Evolution of the Concept

The term "entourage effect" was coined by Mechoulam and Ben-Shabat (1998) in an endocannabinoid context that bears only tangential resemblance to its current popular usage. Their original observation was precise and circumscribed: inactive 2-acyl-glycerol esters—2-linoleoyl-glycerol and 2-palmitoyl-glycerol—that co-occur with 2-arachidonoylglycerol (2-AG) in spleen, brain, and gut potentiated 2-AG's binding at CB receptors, inhibition of adenylate cyclase, and production of the cannabinoid tetrad in vivo, despite having no individual receptor activity. This was an endocannabinoid observation about lipid companions modulating a specific endocannabinoid's activity at its cognate receptors.

Russo's landmark review "Taming THC" (2011, cited over 2,000 times) fundamentally reframed the concept, extending it from endocannabinoid biochemistry to phytocannabinoid-terpenoid synergy. Russo proposed specific terpene-cannabinoid combinations for pain, depression, anxiety, epilepsy, cancer, and methicillin-resistant Staphylococcus aureus, reviewing the therapeutic attributes of eight terpenoids—limonene, myrcene, α-pinene, linalool, β-caryophyllene, caryophyllene oxide, nerolidol, and phytol—and their potential synergies with THC and CBD. The review explicitly cited aromatherapy research, noting that terpenoids "are quite potent, and affect animal and even human behaviour when inhaled from ambient air at serum levels in the single digits ng/mL." This sentence, buried in a cannabis pharmacology

review, is arguably the earliest published acknowledgment of the molecular bridge between aromatherapy and cannabis science that the present review formalizes.

Russo updated the argument in 2019, citing the Pamplona et al. (2018) meta-analysis of 670 patients with treatment-resistant epilepsy, which found that CBD-rich whole-plant cannabis extracts required only 22.5% of the purified CBD dose for equivalent seizure reduction (6.0 versus 25.3 mg/kg/day average dose), accompanied by dramatically fewer adverse events: mild adverse events in 33% versus 76% for purified CBD ($p < 0.001$), and severe adverse events in 7% versus 26% ($p < 0.0001$). While observational and non-blinded, this remains the largest clinical dataset suggesting that co-occurring plant compounds—potentially including terpenes—enhance cannabinoid therapeutic efficacy while reducing toxicity.

B. The Finlay Null Result and Its Consequences

The entourage effect concept faced its most serious empirical challenge with Finlay et al. (2020), who tested α-pinene, β-pinene, β-caryophyllene, linalool, and limonene individually and in mixtures at concentrations up to 10 μM in HEK293 cells stably expressing human CB1 or CB2 receptors. Using radioligand displacement with [^{3}H]CP55,940 and functional cAMP assays, the investigators found no evidence that any terpene displaced the radioligand, altered THC or CBD binding affinity, or modulated cAMP signaling. Their conclusion was unambiguous: “These data do not support the idea that any of the five terpenes tested contribute to a putative entourage effect directly through the cannabinoid receptors.”

This result was widely interpreted—particularly in the popular science press—as disproving the entourage effect entirely. The interpretation was premature for three reasons that subsequent research has clarified. First, the Finlay assays tested orthosteric binding and cAMP modulation, which are standard but not exhaustive measures of receptor engagement. LaVigne et al. (2021) subsequently detected CB1-dependent ERK activation by the same terpenes at higher concentrations using more sensitive assays, demonstrating that terpenes can engage CB1-

dependent signaling cascades without orthosteric displacement. Second, Raz et al. (2023, also published as Revel et al.) detected all 16 cannabis terpenes tested activating CB1 at 10–50% of THC's efficacy using the highly sensitive GIRK (G protein–coupled inwardly rectifying potassium channel) assay in Xenopus oocytes—a system that detects functional receptor activation with greater sensitivity than radioligand binding. Third, and most fundamentally, the Schwarz et al. (2024) identification of adenosine A2a receptors as the primary terpene antinociceptive mechanism revealed that the Finlay study asked the wrong question: terpene analgesia does not require direct CB1 or CB2 activation.

The Finlay result was therefore technically correct—terpenes show negligible direct CB1/CB2 orthosteric binding at the concentrations tested—but the conclusion that the entourage effect was disproven rested on the assumption that such binding was the only possible mechanism. The field's conceptual error was conflating "entourage effect" with "direct cannabinoid receptor modulation," when the actual phenomenon operates through parallel pharmacological pathways.

C. The LaVigne–Schwarz–Spindle Triad: From Tetrad to Human Translation

The experimental resolution of the entourage effect controversy emerged through a remarkably coherent three-study progression from John Streicher's laboratory at the University of Arizona and the Johns Hopkins Behavioral Pharmacology Research Unit.

LaVigne et al. (2021) tested α-humulene, geraniol, linalool, and β-pinene at 200 mg/kg intraperitoneally in CD-1 mice (n = 10–20 per group). All four terpenes produced the full cannabinoid tetrad—analgesia, hypothermia, catalepsy, and hypolocomotion—behaviors classically associated with cannabinoid receptor activation. The CB1 antagonist rimonabant (10 mg/kg) fully blocked terpene-induced tail-flick antinociception but, critically, did not reverse terpene-induced hypothermia, indicating that different tetrad components are mediated by different receptor systems. The A2a antagonist istradefylline blocked terpene-induced catalepsy

specifically. In vitro, all four terpenes activated CB1-dependent ERK signaling. When terpenes were co-administered with the synthetic cannabinoid WIN55,212-2, the combination produced enhanced, prolonged antinociception beyond either compound alone—the first direct experimental demonstration of selective additive effects between terpenes and cannabinoids.

Schwarz et al. (2024) represents the mechanistic breakthrough. Five terpenes (geraniol, linalool, β-pinene, α-humulene, and β-caryophyllene) at 200 mg/kg intraperitoneally produced antinociception roughly equal to 10 mg/kg morphine in chronic constriction injury and chemotherapy-induced peripheral neuropathy models, subsequently confirmed across fibromyalgia and post-operative pain models by Seekins et al. (2025). The adenosine A2a mechanism was established through four converging lines of evidence: (1) systemic administration of the selective A2a antagonist istradefylline (3.2 mg/kg) significantly reduced terpene antinociception; (2) spinal cord–specific CRISPR knockdown of A2a receptors recapitulated the antagonist effect, confirming anatomical specificity; (3) in vitro cAMP modulation assays demonstrated A2a engagement; and (4) in silico molecular docking provided structural plausibility. Crucially, terpenes showed no reward liability in conditioned place preference testing, distinguishing them mechanistically and clinically from opioid analgesics. Low-dose terpene (100 mg/kg) combined with low-dose morphine (3.2 mg/kg) produced synergistic antinociception exceeding either alone, suggesting potential as opioid-sparing agents.

Schwarz et al. (2025) extended these findings, confirming that natural cannabis terpene blends work through A2a, with a critical nuance: purified myrcene but not d-limonene recapitulated antinociceptive effects when administered individually. This terpene-specific finding has direct implications for strain selection and formulation, suggesting that not all terpenes contribute equally to analgesic entourage effects.

Spindle et al. (2024) provided the first controlled human evidence. In a randomized, double-blind, within-subject crossover design with 20 healthy adults, vaporized d-limonene (1, 5, or 15 mg) was co-administered with THC (15 or 30 mg). The highest limonene dose (15 mg) combined with the highest THC dose (30 mg) significantly reduced self-reported ratings of

"anxious/nervous" and "paranoid" compared with THC alone, in an orderly dose-response pattern. The selectivity of this effect is its most striking feature: limonene did not alter any other measured THC effect—not the subjective "high," not cognitive impairment, not vital signs, and not THC pharmacokinetics. Limonene administered alone produced no pharmacodynamic effects at any dose. This pattern is consistent with A2a-mediated anxiolysis operating independently of CB1-mediated intoxication, precisely the kind of selective modulation that would be expected if terpene-cannabinoid interactions operate through parallel receptor systems rather than direct cannabinoid receptor modulation.

Kumar et al. (2025) from the same Johns Hopkins laboratory provides an important negative counterpoint: α-pinene co-vaporized with THC in 19 healthy adults did not attenuate THC-induced memory impairment. This null result is informative rather than damaging to the entourage effect framework, suggesting that either the dose was insufficient, memory represents a different pharmacological target than anxiety, or α-pinene's AChE inhibition mechanism does not interact with THC's memory effects at the concentrations achieved through vaporization. The contrast between the Spindle positive result (limonene–anxiety) and the Kumar negative result (α-pinene–memory) underscores that the entourage effect is not a monolithic phenomenon but a set of terpene-specific, endpoint-specific interactions that must be characterized individually.

D. The Dose Gap and Its Emerging Resolution

The most persistent criticism of the entourage effect concerns the dose gap between preclinical and clinical contexts. Raz et al. (2024, 2025) explicitly acknowledged that the terpene doses producing robust effects in mouse models (200 mg/kg intraperitoneally) are "substantially higher than those attainable through cannabis flower consumption (≤20 mg total)," potentially leading to overestimation of therapeutic effects. A 70-kg human consuming cannabis flower inhales

roughly 1–5 mg of any individual terpene per session—roughly 0.01–0.07 mg/kg, representing a gap of approximately three to four orders of magnitude from the animal dosing paradigm.

Three lines of evidence have partially resolved this gap. First, the Raz et al. (2023) GIRK assay data demonstrated that terpene-mediated CB1 amplification of THC activity occurs at concentrations as low as 0.001–0.01 μM, at terpene-to-THC ratios matching those found naturally in the cannabis plant. This is a fundamentally different pharmacological principle from high-dose monotherapy: rather than producing independent receptor activation at pharmacological concentrations, terpenes synergistically amplify cannabinoid signaling at physiologically realistic concentrations through allosteric or functional interactions. If confirmed in vivo, this mechanism would operate within the concentration range achievable through normal cannabis consumption.

Second, the identification of A2a as an independent analgesic pathway (Schwarz et al., 2024) introduced a mechanism that, while currently demonstrated only at high doses in mice, operates through a receptor system with well-characterized pharmacology amenable to optimization. The failure of terpene inhalation (as opposed to intraperitoneal injection) to produce antinociception in the Schwarz mouse model represents a significant translational limitation that must be addressed before clinical relevance can be established.

Third, the pre-loading paradigm—using external botanical terpene sources (essential oil inhalation, forest exposure) to supplement cannabis-derived terpenes—offers a practical approach to increasing total terpene exposure above cannabis-only levels. The pharmacokinetic data reviewed in Section 2 establish that inhaled terpenes achieve rapid systemic absorption (60–65% pulmonary uptake) with measurable brain concentrations within 30 minutes (Satou et al., 2016). Critically, Satou demonstrated that α-pinene brain concentrations were two-fold higher when inhaled as a mixed essential oil blend versus single-compound inhalation, suggesting that multi-compound delivery—the default mode of both aromatherapy and cannabis consumption—enhances individual terpene bioavailability through mechanisms not yet characterized.

The Spindle et al. (2024) human result provides the strongest evidence that clinically meaningful terpene-cannabinoid interactions can occur at realistic doses: just 15 mg of vaporized limonene produced selective anxiolysis. This dose, while administered via a controlled vaporization system delivering pure limonene, falls within the range potentially achievable through combined cannabis plus supplementary terpene exposure. The pre-loading concept would further extend this by establishing terpene receptor occupancy before cannabis consumption, a temporal sequence consistent with the allosteric priming hypothesis detailed in Section 6.

E. Reconceptualizing the Entourage Effect Within the Unified Framework

The evidence reviewed in this section supports a fundamental reconceptualization of the entourage effect. The original framing—which implied that terpenes enhance cannabinoid effects through direct modulation of CB1 or CB2 receptors—has been empirically tested and found wanting at the level of orthosteric receptor binding. The replacement framework is both more complex and more clinically promising: terpenes operate through parallel pharmacological pathways, primarily adenosine A2a receptors for analgesia and potentially through allosteric CB1 modulation at plant-relevant concentrations, to produce selective effects that complement rather than simply amplify cannabinoid activity.

This reconceptualization has three implications for the unified terpene pharmacology framework. First, it dissolves the apparent contradiction between the Finlay null result and the clinical reality reported by patients and practitioners who observe differences between cannabis chemotypes with different terpene profiles. The effects are real; the mechanism was incorrectly assumed to require direct cannabinoid receptor binding. Second, it establishes that terpene-cannabinoid interactions are not monolithic but terpene-specific and endpoint-specific—limonene selectively reduces anxiety (Spindle et al., 2024), myrcene but not limonene produces A2a-mediated analgesia (Schwarz et al., 2025), and α-pinene does not attenuate THC-induced

memory impairment at vaporized doses (Kumar et al., 2025). Third, and most relevant to the present review, it reveals that the entourage effect operates through the same receptor systems—A2a, GABA-A, TRP channels, serotonergic pathways—that mediate the clinical effects documented in the aromatherapy and forest medicine literatures reviewed in Sections 3 and 5. The entourage effect is not a cannabis-specific phenomenon requiring cannabis-specific mechanisms; it is a manifestation of the polypharmacological architecture of terpenes operating within a particular botanical context.

Table 4. Chronological progression of entourage effect evidence from concept to mechanism. The Finlay (2020) null result for direct CB1/CB2 binding was resolved by the identification of adenosine A2a as the primary terpene antinociceptive mechanism (Schwarz et al., 2024) and detection of allosteric CB1 modulation at plant-relevant concentrations (Raz et al., 2023). DB, double-blind; CCI, chronic constriction injury; CIPN, chemotherapy-induced peripheral neuropathy; GIRK, G protein–coupled inwardly rectifying potassium channel; AE, adverse event; BDZ, benzodiazepine.

Year	Study	Model / Design	Key Finding	Mechanism Identified	Dose / Concentration
1998	Mechoulam & Ben-Shabat (Eur. J. Pharmacol.)	In vitro / in vivo	Inactive 2-acyl-glycerol esters potentiated 2-AG at CB receptors; coined "entourage effect"	Endocannabinoid system	Endogenous concentrations
2011	Russo (Br. J. Pharmacol.) "Taming THC"	Review / theoretical	Extended entourage to phytocannabinoid-terpenoid synergy; proposed 8 terpene-cannabinoid combinations for specific conditions	Multiple proposed (GABA-A, 5-HT, AChE, CB2)	Theoretical

2018 Pamplona et al. (Front. Neurol.) Meta-analysis, 670 epilepsy patients CBD-rich extracts 4× more potent than purified CBD (6.0 vs 25.3 mg/kg/day); fewer AEs (33% vs 76%) Unspecified multi-compound synergy Clinical doses

2020 Finlay et al. (Front. Pharmacol.) HEK293, CB1/CB2, radioligand + cAMP No terpene displaced CP55,940 or modulated cAMP at ≤10 μM; no direct CB1/CB2 orthosteric binding NULL RESULT for CB1/CB2 ≤10 μM in vitro

2020 Ferber et al. (Curr. Neuropharmacol.) Review Limonene anxiolysis NOT blocked by flumazenil; proposed non-BDZ mechanism for terpene-cannabinoid mood modulation Non-GABA-A (later identified as A2a) 1% inhaled

2021 LaVigne et al. (Sci. Rep./Nature) CD-1 mice, n=10–20/group 4 terpenes (200 mg/kg IP) produced full cannabinoid tetrad; rimonabant blocked analgesia; istradefylline blocked catalepsy; terpene + WIN55 = enhanced antinociception CB1-dependent ERK + A2a (catalepsy) 200 mg/kg IP

2023 Raz et al. (Biochem. Pharmacol.) Xenopus oocytes, GIRK assay 16 terpenes each activated CB1 at 10–50% THC efficacy; major amplification at plant-ratio concentrations (0.001–0.01 μM) CB1 allosteric/functional 0.001–0.01 μM (plant-relevant)

2024 Schwarz et al. (PAIN) CCI, CIPN, fibromyalgia models; CRISPR 5 terpenes (200 mg/kg) ≈ 10 mg/kg morphine; A2a antagonist + spinal CRISPR confirmed mechanism; no reward liability Adenosine A2a (primary) 200 mg/kg IP

2024 Spindle et al. (Drug Alcohol Depend.) 20 humans, DB crossover Limonene (15 mg) + THC (30 mg) selectively reduced anxiety/paranoia vs THC alone; no other

THC effects altered A2a-mediated (inferred)

15 mg vaporized (human)

2025 Kumar et al. (Med. Cannabis Cannabinoids) 19 humans, DB crossover

α-Pinene + THC did NOT attenuate memory impairment—important negative result

Insufficient dose or wrong endpoint Vaporized (human)

2025 Schwarz et al. (Pharmacol. Rep.) Post-operative pain, CRISPR

Confirmed across 4 pain types; natural cannabis terpene blends work via A2a; myrcene but NOT limonene effective alone A2a confirmed; terpene-specific

200 mg/kg IP ------ ---------- ---------- ---------- ---------- ---------- ---------- ----------

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Figure 3. The dose gap and the pre-loading resolution. Panel A: Logarithmic comparison of terpene doses/concentrations across experimental contexts, from LaVigne/Schwarz animal studies (200 mg/kg IP) through Finlay in vitro testing (≤10 μM, null result for CB1/CB2) to Raz GIRK assay (0.001–0.01 μM, plant-relevant CB1 activation) and Spindle human trial (15 mg vaporized, selective anxiolysis). The ∼1,000× dose gap between animal models and human cannabis consumption is annotated. Panel B: Three converging pathways toward resolution—in vitro CB1 synergy at plant concentrations (Raz et al., 2023), A2a as an independent non-cannabinoid analgesic mechanism (Schwarz et al., 2024), and the pre-loading paradigm using external botanical terpene sources to supplement cannabis-derived terpenes. Panel C: Conceptual evolution from Mechoulam's original endocannabinoid entourage (1998) through the Russo terpene extension (2011), the Finlay null result crisis (2020), and resolution via A2a mechanism identification and human validation (2024–2025).

Section 4 word count: ~3,800 words | Cumulative (Sections 1–4): ~14,400 words of target 15,000–25,000 total manuscript

Table 4. Chronological progression of entourage effect evidence from concept to mechanism. The Finlay (2020) null result for direct CB1/CB2 binding was resolved by the identification of adenosine A2a as the primary terpene antinociceptive mechanism (Schwarz et al., 2024) and detection of allosteric CB1 modulation at plant-relevant concentrations (Raz et al., 2023). DB, double-blind; CCI, chronic constriction injury; CIPN, chemotherapy-induced peripheral neuropathy; GIRK, G protein–coupled inwardly rectifying potassium channel; AE, adverse event; BDZ, benzodiazepine.

Year	Study	Model / Design	Key Finding	Mechanism Identified	Dose / Concentration
1998	Mechoulam & Ben-Shabat (Eur. J. Pharmacol.)	In vitro / in vivo	Inactive 2-acyl-glycerol esters potentiated 2-AG at CB receptors; coined "entourage effect"	Endocannabinoid system	Endogenous concentrations
2011	Russo (Br. J. Pharmacol.) "Taming THC"	Review / theoretical	Extended entourage to phytocannabinoid-terpenoid synergy; proposed 8 terpene-cannabinoid combinations for specific conditions	Multiple proposed (GABA-A, 5-HT, AChE, CB2)	Theoretical
2018	Pamplona et al. (Front. Neurol.)	Meta-analysis, 670 epilepsy patients	CBD-rich extracts 4× more potent than purified CBD (6.0 vs 25.3 mg/kg/day); fewer AEs (33% vs 76%)	Unspecified multi-compound synergy	Clinical doses
2020	Finlay et al. (Front. Pharmacol.)	HEK293, CB1/CB2, radioligand + cAMP	No terpene displaced CP55,940 or modulated cAMP at ≤10 μM; no direct CB1/CB2 orthosteric binding	NULL RESULT for CB1/CB2	≤10 μM in vitro
2020	Ferber et al. (Curr. Neuropharmacol.)	Review	Limonene anxiolysis NOT blocked by flumazenil; proposed non-BDZ mechanism for terpene-cannabinoid mood modulation	Non-GABA-A (later identified as A2a)	1% inhaled

2021	LaVigne et al. (Sci. Rep./Nature)	CD-1 mice, n=10–20/group	4 terpenes (200 mg/kg IP) produced full cannabinoid tetrad; rimonabant blocked analgesia; istradefylline blocked catalepsy; terpene + WIN55 = enhanced antinociception	CB1-dependent ERK + A2a (catalepsy)	200 mg/kg IP
2023	Raz et al. (Biochem. Pharmacol.)	Xenopus oocytes, GIRK assay	16 terpenes each activated CB1 at 10–50% THC efficacy; major amplification at plant-ratio concentrations (0.001–0.01 μM)	CB1 allosteric/functional	0.001–0.01 μM (plant-relevant)
2024	Schwarz et al. (PAIN)	CCI, CIPN, fibromyalgia models; CRISPR	5 terpenes (200 mg/kg) ≈ 10 mg/kg morphine; A2a antagonist + spinal CRISPR confirmed mechanism; no reward liability	Adenosine A2a (primary)	200 mg/kg IP
2024	Spindle et al. (Drug Alcohol Depend.)	20 humans, DB crossover	Limonene (15 mg) + THC (30 mg) selectively reduced anxiety/paranoia vs THC alone; no other THC effects altered	A2a-mediated (inferred)	15 mg vaporized (human)
2025	Kumar et al. (Med. Cannabis Cannabinoids)	19 humans, DB crossover	α-Pinene + THC did NOT attenuate memory impairment—important negative result	Insufficient dose or wrong endpoint	Vaporized (human)
2025	Schwarz et al. (Pharmacol. Rep.)	Post-operative pain, CRISPR	Confirmed across 4 pain types; natural cannabis terpene blends work via A2a; myrcene but NOT limonene effective alone	A2a confirmed; terpene-specific	200 mg/kg IP

Section 5: Forest Bathing Immunology

Pharmacological Reviews (ASPET)

Prepared for submission to Pharmacological Reviews (ASPET)

V. Forest Bathing Immunology: Phytoncides as the Mechanism Behind Shinrin-Yoku

A. Qing Li's NK Cell Research Program

The forest bathing (shinrin-yoku) research program initiated by Qing Li at Nippon Medical School represents the third major body of evidence demonstrating that inhaled terpenes produce measurable physiological effects in humans. The significance of this literature for the unified terpene pharmacology framework lies not in its novelty—the effects of forest environments on human health have been observed for millennia—but in its progressive isolation of terpenes as the active variable within the complex multi-sensory forest environment.

Li's foundational study (2007) exposed 12 healthy males to a three-day, two-night forest trip and measured natural killer cell parameters before, during, and after the exposure. NK cell activity increased approximately 50%, with 11 of 12 subjects showing elevated activity. Significant increases were documented in NK cell numbers and in intracellular concentrations of the cytolytic proteins perforin, granulysin, and granzymes A and B—markers of enhanced anti-tumor immune surveillance. The controlled comparison study (Li et al., 2008a) employed the same 12 male subjects in a forest visit versus a Nagoya city tourist trip, finding that the forest exposure increased NK activity and anti-cancer proteins while the city trip produced no immunological changes. Critically, α-pinene and β-pinene were detected in forest air samples but were absent from city air, implicating volatile terpenes as candidate mediators. Li et al. (2008b) confirmed these results in 13 female nurses using the same protocol, establishing cross-sex reproducibility.

The duration of immune enhancement proved remarkably persistent. Li's 2010 comprehensive review synthesized data showing effects lasting more than seven days after forest exposure and concluded that "a forest bathing trip once a month would enable individuals to maintain a higher level of NK activity"—a practical recommendation grounded in the pharmacokinetic reality that monoterpenes exhibit deep-compartment half-lives of up to 11.6 hours (Section 2), with adipose tissue serving as a slow-release reservoir.

B. The Hotel Room Experiment: Isolating the Phytoncide Variable

The methodological challenge inherent in forest bathing research is the confounding of terpene inhalation with the full sensory ecology of the forest environment—visual green exposure, natural soundscapes, reduced urban stressors, physical activity in natural settings, and social context. Li et al. (2009) addressed this confound directly in what remains the most important single study in the forest medicine literature.

Twelve healthy male subjects stayed overnight in an urban hotel room in which hinoki cypress (Chamaecyparis obtusa) essential oil was vaporized via a humidifier. The essential oil contained the same monoterpenes detected in forest air—primarily α-pinene, with sabinene, limonene, and other constituents. NK cell activity, anti-cancer protein expression (perforin, granzyme A, granulysin), and urinary catecholamine levels were measured before and after the exposure. The results replicated the forest bathing immunological profile: NK activity and intracellular cytolytic proteins increased significantly, while urinary adrenaline and noradrenaline decreased.

This experiment accomplished what no forest field study could: it isolated the volatile terpene variable from visual, auditory, and psychological confounders. The subjects were in an urban hotel room—no green views, no birdsong, no forest soundscape—yet their immune parameters shifted in the same direction and magnitude as subjects immersed in actual forests. The implication is unambiguous: airborne terpenes are sufficient to produce the NK cell

enhancement documented in forest bathing research. This finding is directly relevant to the aromatherapy literature (Section 3) and the pre-loading paradigm (Section 4), as it demonstrates that essential oil diffusion in an indoor environment can replicate immunological effects attributed to forest exposure.

C. Mechanistic Studies: From Correlation to Causation

The in vitro phytoncide study (Li et al., 2006) preceded the forest field studies and established the direct biological plausibility of terpene-mediated NK cell activation. α-Pinene, 1,8-cineole, and d-limonene were incubated with NK-92MI cells for 48 to 144 hours, producing dose-dependent increases in cytolytic activity and intracellular concentrations of perforin, granzyme A, and granulysin. This demonstrated that terpenes can activate NK cells without any intermediary—no olfactory processing, no stress reduction, no psychological mediation—through direct molecular interaction with immune cells.

Kim et al. (2021) extended this work by identifying the intracellular signaling cascade: α-pinene at 100 μM increased perforin, granzyme B, CD56, and CD107a expression in NK-92mi cells via ERK/AKT phosphorylation. The in vivo component of this study demonstrated that α-pinene treatment inhibited CT-26 colon tumor growth in mice, providing the first direct evidence that a forest air terpene can produce anti-tumor effects through NK cell activation in a living organism.

The meta-analytic synthesis by Piva et al. (2024) pooled data across the phytoncide literature and calculated an NK cell activation effect size of 2.50 (95% CI: 1.94–3.05, $p < 0.05$, $I^2 = 50.47\%$). This is a very large effect by conventional standards, though the moderate heterogeneity ($I^2 = 50\%$) reflects the diversity of protocols, from controlled in vitro exposures to multi-day forest immersions with multiple confounders. The consistency of direction across this heterogeneous literature—every study reporting NK enhancement—strengthens the case that the

underlying biological signal is genuine, even if the precise effect magnitude varies with methodology.

D. Beyond Immunity: Serotonin, Oxytocin, and Clinical Populations

Li's 2022–2025 research program has expanded the documented effects of forest bathing beyond the NK cell–cancer surveillance axis into neuropsychiatric and respiratory domains. Li et al. (2022) provided the first measurement of blood serotonin in a forest bathing context, demonstrating significant increases accompanied by reduced depression scores. Li et al. (2025a, Diseases) extended this to a clinical population: 31 women with depression showed decreased Self-rating Depression Scale scores alongside increased serotonin, oxytocin, and insulin-like growth factor 1 (IGF-1), with effects persisting one week after exposure. The inclusion of oxytocin—a neuropeptide associated with social bonding, trust, and anxiolysis—is novel and suggests that forest terpene exposure engages broader neuroendocrine pathways than previously appreciated.

Li et al. (2025b, J. Occup. Health) documented the first application of forest bathing in a respiratory disease population: 30 males with chronic obstructive pulmonary disease risk factors showed reduced inflammatory markers and improved peripheral oxygen saturation (SpO_2) after forest bathing in a randomized crossover design. This connects the forest medicine literature to the Section 2 finding that terpenes achieve 60–65% pulmonary uptake—the same inhalation route delivering the same molecules to the same respiratory epithelium.

E. The Levy Null Result: Methodological Rigor Reveals Complexity

The most methodologically rigorous study in the forest medicine literature—and the one most challenging to the phytoncide hypothesis—is Levy et al. (2025, Environ. Res. 276:121482). This double-blind, randomized crossover study filtered forest air to remove terpenes and compared psychophysiological and immunological outcomes between filtered and unfiltered 60-minute

forest air sessions. The investigators found no consistent differences between conditions across their outcome measures.

This null result must be interpreted carefully rather than dismissed or overinterpreted. The 60-minute exposure duration was substantially shorter than the multi-day protocols used in Li's positive studies (two to three nights), and shorter even than the overnight hotel room exposure that successfully replicated NK enhancement. The pharmacokinetic data reviewed in Section 2 suggest that α-pinene requires approximately 30 minutes to reach peak brain concentrations, with deep-compartment redistribution continuing for hours. A 60-minute session may simply be too brief for the cumulative terpene dose to reach the threshold required for immunological effects—consistent with the dose–response relationship established across the Li program, where longer exposures produced larger and more persistent effects.

The Levy result also raises the possibility that multi-sensory integration—the congruence of visual, auditory, olfactory, and proprioceptive forest stimuli—may be necessary for the full magnitude of forest bathing effects, even if terpenes are a necessary component. The hotel room experiment (Li et al., 2009) demonstrated that terpenes are sufficient for NK enhancement, but the Levy result suggests they may not be sufficient within a 60-minute window when other forest stimuli are held constant. This temporal threshold hypothesis is testable and represents a priority for future research.

F. The Concentration Gradient: Same Molecules, Five Orders of Magnitude

Forest air contains total terpenes at concentrations of 3.5–56.0 μg/m³, with α-pinene reaching up to 13.05 μg/m³ in coniferous forests and limonene at 0.81–8.10 μg/m³. These concentrations are one thousand to one hundred thousand times lower than those achieved through essential oil diffusion or cannabis vaporization. Yet biological effects remain demonstrable at forest air concentrations, potentially operating through olfactory-limbic signaling rather than systemic receptor engagement at pharmacological concentrations.

The NHANES 2013–2014 population study (n = 2,213) documented baseline serum terpene levels in the general United States population: limonene at approximately 1,200 ng/L and α-pinene at approximately 76 ng/L, with dietary citrus consumption increasing serum limonene by 82.1%. These baseline levels establish that the human population maintains constitutive low-level terpene exposure through diet and environment, providing a background against which acute exposures—whether from forest air, essential oil diffusion, or cannabis consumption—produce incremental pharmacological effects.

The concentration gradient spanning these contexts—forest air, essential oil diffusion, direct nasal inhalation, cannabis vaporization—represents the same molecules engaging the same receptor systems across five orders of magnitude (Figure 2, Section 2). Forest medicine research demonstrates that biological responses occur even at the lowest end of this gradient, while the aromatherapy and cannabis literatures document effects at progressively higher concentrations. The unified framework proposes that these represent a continuous dose–response relationship rather than qualitatively different phenomena, with the olfactory-limbic pathway providing a mechanism through which even low terpene concentrations can produce neurological effects by bypassing the blood-brain barrier entirely.

G. Integrating Forest Medicine into the Unified Framework

The forest bathing literature contributes three essential elements to the unified terpene pharmacology framework. First, it provides the strongest evidence for immune modulation by inhaled terpenes, with the hotel room experiment isolating the phytoncide variable from environmental confounders and the Piva meta-analysis confirming a very large pooled effect size for NK cell activation. Second, it demonstrates that the same terpenes studied in cannabis pharmacology and aromatherapy—α-pinene, limonene, 1,8-cineole—produce measurable physiological effects through inhalation, establishing a third independent evidence stream supporting the biological activity of these molecules regardless of botanical source. Third, the

expansion into serotonergic, oxytocinergic, and respiratory endpoints (Li et al., 2022–2025) broadens the documented pharmacological scope of inhaled terpenes beyond the immune and neurological effects characterized in the aromatherapy and cannabis literatures.

The Levy null result serves as a methodological corrective, demonstrating that the field's most rigorous experimental design—double-blind air filtration—did not replicate terpene-mediated effects within a 60-minute exposure window. This finding disciplines the framework by establishing that forest terpene effects are not instantaneous or inevitable but depend on exposure duration, cumulative dose, and potentially multi-sensory context. The temporal threshold hypothesis—that meaningful terpene exposure requires longer than 60 minutes of forest-level concentrations—is consistent with the pharmacokinetic evidence of Section 2 and represents a testable prediction that connects forest medicine, aromatherapy (where clinical effects emerge with exposures of 30–90 minutes of essential oil diffusion at much higher concentrations), and the pre-loading paradigm proposed for cannabis terpene augmentation.

Table 5. Chronological progression of forest bathing and phytoncide immunology evidence. The Li hotel room experiment (2009) represents the strongest evidence isolating the phytoncide variable from confounding multi-sensory forest stimuli. The Levy et al. (2025) null result from a double-blind filtered air design suggests that 60-minute exposures may be insufficient to replicate effects documented in multi-day studies. NK, natural killer; SDS, Self-rating Depression Scale; DB, double-blind; IGF-1, insulin-like growth factor 1; SpO_2, peripheral oxygen saturation.

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--- Year Study Design / Sample

Key Finding

Methodological Significance Limitation / Note

2006 Li et al. (Immunopharmacol. Immunotoxicol.) NK-92MI cells, in vitro
α-Pinene, 1,8-cineole, d-limonene dose-dependently increased cytolytic activity, perforin, granzyme A, granulysin (48–144 h incubation) In vitro: direct phytoncide–NK interaction Isolated terpene variable

2007 Li et al. (Int. J. Immunopathol. Pharmacol.) 12 males, 3-day forest trip
NK activity increased ~50%; 11/12 subjects showed elevated NK; increased perforin, granulysin, granzymes A/B; effects lasted >7 days In vivo forest: multi-sensory exposure
No city control group

2008a Li et al. (Int. J. Immunopathol. Pharmacol.) 12 males, forest vs. Nagoya city
Forest trip increased NK activity and anti-cancer proteins; city trip produced no changes; α-pinene/β-pinene detected in forest air, absent in city Controlled comparison: forest vs. city First controlled design

2008b Li et al. (J. Biol. Regul. Homeost. Agents) 13 female nurses
Confirmed forest NK enhancement across sex; same protocol as 2008a
Cross-sex replication Female cohort

2009 Li et al. (Int. J. Immunopathol. Pharmacol.) 12 males, urban hotel room
Hinoki cypress oil vaporized via humidifier overnight; NK activity, anti-cancer proteins increased; catecholamines decreased—replicating forest effects indoors STRONGEST: isolated phytoncide variable in controlled setting Indoor replication

2021 Kim et al. (Int. J. Mol. Sci.) NK-92mi cells + CT-26 in vivo
α-Pinene (100 μM) increased perforin, granzyme B, CD56, CD107a via ERK/AKT; inhibited tumor growth in vivo Mechanistic: ERK/AKT signaling identified Mouse tumor model

2022 Li et al. (Environ. Health Prev. Med.) Forest bathing + blood serotonin
First serotonin measurement in forest bathing context; significant increase with reduced depression scores Serotonergic pathway added
First 5-HT data

2024 Piva et al. (meta-analysis) Pooled phytoncide studies NK cell activation effect size 2.50 (95% CI: 1.94–3.05, $p<0.05$, $I^2=50.47\%$)—very large effect Meta-analytic confirmation Moderate heterogeneity

2025 Levy et al. (Environ. Res.) DB randomized crossover, filtered vs. unfiltered forest air No consistent differences in psychophysiological or immunological outcomes between filtered and unfiltered 60-min forest air—IMPORTANT NULL RESULT NULL: 60 min insufficient; multi-sensory integration may be required Most rigorous design

2025 Li et al. (J. Occup. Health) 30 males with COPD risk, randomized crossover Reduced inflammatory markers, improved SpO_2; first COPD application of forest bathing Anti-inflammatory in respiratory disease Clinical population

2025 Li et al. (Diseases) 31 women with depression Decreased SDS scores; increased serotonin, oxytocin, IGF-1; effects persisting one week Neuroendocrine: serotonin + oxytocin + IGF-1 Psychiatric population ---

Section 5 word count: ~3,100 words | Cumulative (Sections 1–5): ~16,400 words of target 15,000–25,000 total manuscript

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Year	Study	Design / Sample	Key Finding	Methodological Significance	Limitation / Note
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2007	Li et al. (Int. J. Immunopathol. Pharmacol.)	12 males, 3-day forest trip	NK activity increased ~50%; 11/12 subjects showed elevated NK; increased perforin, granulysin, granzymes A/B; effects lasted >7 days	In vivo forest: multi-sensory exposure	No city control group
2008a	Li et al. (Int. J. Immunopathol. Pharmacol.)	12 males, forest vs. Nagoya city	Forest trip increased NK activity and anti-cancer proteins; city trip produced no changes; α-pinene/β-pinene detected in forest air, absent in city	Controlled comparison: forest vs. city	First controlled design
2008b	Li et al. (J. Biol. Regul. Homeost. Agents)	13 female nurses	Confirmed forest NK enhancement across sex; same protocol as 2008a	Cross-sex replication	Female cohort

2009	Li et al. (Int. J. Immunopathol. Pharmacol.)	12 males, urban hotel room	Hinoki cypress oil vaporized via humidifier overnight; NK activity, anti-cancer proteins increased; catecholamines decreased—replicating forest effects indoors	STRONGEST: isolated phytoncide variable in controlled setting	Indoor replication
2021	Kim et al. (Int. J. Mol. Sci.)	NK-92mi cells + CT-26 in vivo	α-Pinene (100 μM) increased perforin, granzyme B, CD56, CD107a via ERK/AKT; inhibited tumor growth in vivo	Mechanistic: ERK/AKT signaling identified	Mouse tumor model
2022	Li et al. (Environ. Health Prev. Med.)	Forest bathing + blood serotonin	First serotonin measurement in forest bathing context; significant increase with reduced depression scores	Serotonergic pathway added	First 5-HT data
2024	Piva et al. (meta-analysis)	Pooled phytoncide studies	NK cell activation effect size 2.50 (95% CI: 1.94–3.05, $p<0.05$, $I^2=50.47\%$)—very large effect	Meta-analytic confirmation	Moderate heterogeneity
2025	Levy et al. (Environ. Res.)	DB randomized crossover, filtered vs. unfiltered forest air	No consistent differences in psychophysiological or immunological outcomes between filtered and unfiltered 60-min forest air—IMPORTANT NULL RESULT	NULL: 60 min insufficient; multi-sensory integration may be required	Most rigorous design

2025	Li et al. (J. Occup. Health)	30 males with COPD risk, randomized crossover	Reduced inflammatory markers, improved SpO_2; first COPD application of forest bathing	Anti-inflammatory in respiratory disease	Clinical population
2025	Li et al. (Diseases)	31 women with depression	Decreased SDS scores; increased serotonin, oxytocin, IGF-1; effects persisting one week	Neuroendocrine: serotonin + oxytocin + IGF-1	Psychiatric population

Section 6: The A2a-CB1 Heteromer System

Pharmacological Reviews (ASPET)

Prepared for submission to Pharmacological Reviews (ASPET)

VI. The A2a-CB1 Heteromer System Provides the Molecular Substrate for Terpene-Cannabinoid Crosstalk

A. Architecture and Location of the Heteromer

The identification of adenosine A2a receptors as the primary mechanism of terpene antinociception (Schwarz et al., 2024; Section 4) raises a fundamental question: if terpenes act through A2a and cannabinoids act through CB1, how do these parallel pathways interact to produce the selective modulation documented in human studies? The answer lies in the A2a-CB1 receptor heteromer—a physical complex between A2a and CB1 receptors that creates a molecular interface for direct crosstalk between terpene-responsive and cannabinoid-responsive signaling systems.

Moreno et al. (2018) provided the definitive mapping of A2a-CB1 heteromer distribution in the dorsal striatum using in situ proximity ligation assay combined with conditional CB1 knockout mice. The central finding was one of striking selectivity: A2a-CB1 heteromers are essentially absent from corticostriatal projections and striatonigral (direct pathway) neurons but are largely present in striatopallidal (indirect pathway) neurons. This pathway-specific distribution has immediate functional implications. Co-stimulation of both receptors within the heteromer produces cross-antagonism—strongly reduced ERK and Akt phosphorylation—meaning that simultaneous A2a and CB1 activation does not produce additive signaling but rather mutual attenuation. The heteromer also exhibits altered G-protein coupling, selectively engaging Gq proteins rather than the individual protomers' canonical Gs (A2a) or Gi (CB1) coupling.

Kőfalvi et al. (2020) established the quaternary structure as an $(A2a)_2$-$(CB1)_2$ heterotetramer. The long A2a C-terminus proved critical, determining constitutive adenylyl cyclase activation via Gs. Within this complex, G-protein coupling maintains specificity: the A2a agonist NECA signals exclusively through Gs (EC_{50} = 0.35 ± 0.08 μM) while the CB1 agonist CP55940 signals exclusively through Gi (EC_{50} = 1.05 ± 0.39 μM). The critical mechanistic insight was that CB1 agonist–mediated inhibition of striatal glutamate release—one of the most well-established cannabinoid effects in the brain—primarily occurs not through independent CB1 signaling but through counteraction of A2a's constitutive adenylyl cyclase activation within the heteromer. This means the canonical cannabinoid effect of reducing excitatory neurotransmission mechanistically depends on the A2a-CB1 heteromer architecture.

B. Hippocampal Heteromers and CBD's A2a-Dependent Mechanism

Aso et al. (2019) extended the heteromer mapping from striatum to hippocampus, revealing A2a-CB1 heteromers at presynaptic terminals in CA1 neurons. The functional experiment was elegant: CBD was found to blunt THC-induced cognitive impairment in an A2a-dependent manner. The specificity of this effect provides the strongest evidence for brain region–dependent functional selectivity of the heteromer: CBD was not capable of modifying motor functions or anxiety through this mechanism—only cognition. This dissociation between cognitive and motor/anxiolytic effects maps directly onto the anatomical distribution: hippocampal heteromers mediate cognitive effects while striatal heteromers mediate motor effects.

The molecular mechanism by which CBD achieves this modulation was subsequently clarified by Sánchez-Fernández et al. (2024), who demonstrated that CBD acts as a negative allosteric modulator of A2a receptors, decreasing agonist (CGS21680) potency by 3.5- to 5-fold without acting as an agonist itself. Raïch et al. (2023) confirmed quantitatively that CBD at 200 nM decreased A2a-downstream ERK phosphorylation by approximately 40%. Together, these findings provide a complete mechanistic chain: CBD modulates THC's cognitive effects by

negatively modulating A2a within the heteromer complex, reducing the A2a-mediated component of CB1 signaling at hippocampal synapses.

C. A2a Antagonism Prevents Cannabinoid-Induced Memory Deficits

Mouro et al. (2017, 2019) provided complementary evidence from a chronic exposure paradigm. Chronic istradefylline (A2a antagonist, 3 mg/kg over 28 days) reverted recognition memory deficits induced by chronic WIN55,212-2 (synthetic CB1 agonist, 1 mg/kg over 28 days). At the synaptic level, WIN at 300 nM impaired hippocampal CA1 long-term potentiation—the electrophysiological correlate of memory formation—and the A2a antagonist SCH 58261 at 100 nM partially rescued this impairment. Importantly, chronic administration of neither agent altered A2a or CB1 receptor density in hippocampus or prefrontal cortex, indicating that the rescue occurred through functional modulation of the existing heteromer population rather than receptor upregulation or downregulation.

Chiodi et al. (2016) approached the same question from the opposite direction, using transgenic rats that overexpress A2a receptors. In these animals, CB1 agonist effects were significantly reduced—a gain-of-function demonstration that increasing A2a expression attenuates CB1-mediated signaling, consistent with the cross-antagonism mechanism documented by Moreno et al. (2018). Carlin et al. (2018) added a further connection to the terpene literature by demonstrating that A2a agonism causes hypothermia via peripheral receptors, linking the A2a pathway to the hypothermia component of the cannabinoid tetrad that LaVigne et al. (2021) found was CB1-independent (Section 4).

D. The Pre-Loading Mechanism at the Heteromer

The heteromer pharmacology described above provides a specific, testable molecular model for how terpene pre-loading would modulate subsequent cannabis effects. Based on the established signaling properties of the A2a-CB1 heterotetramer, activating A2a first—via limonene or other

A2a agonist terpenes delivered through essential oil diffusion—would initiate a defined sequence of events at the heteromer: (1) increased adenylyl cyclase activity through Gs coupling at the A2a protomer; (2) establishment of a Type I allosteric state (Ferré et al., 2023) that reduces subsequent CB1 Gi-mediated signaling efficiency; (3) at corticostriatal terminals, reduced THC-mediated inhibition of glutamate release through interference with A2a's constitutive adenylyl cyclase activation; and (4) at postsynaptic striatopallidal neurons, cross-antagonism producing reduced ERK and Akt phosphorylation relative to CB1 activation alone.

The net predicted effect is selective attenuation of certain CB1-mediated effects—specifically those mediated through the heteromer in striatopallidal and hippocampal neurons—while leaving CB1 effects mediated through homomeric receptors in other brain regions unaffected. This selectivity is precisely what the Spindle et al. (2024) human data showed: limonene reduced THC-induced anxiety and paranoia without altering the subjective high, cognitive effects, vital signs, or THC pharmacokinetics. The heteromer model explains this pattern: anxiety modulation occurs through A2a-containing circuits while the euphoric and cognitive effects of THC are mediated through CB1 populations that are not heteromerized with A2a.

Ferré et al. (2023) formalized three types of allosteric interactions within receptor heteromers that provide the theoretical framework for predicting pre-loading outcomes. Type I interactions are ligand-dependent: binding of an agonist at one protomer changes the binding affinity or signaling efficacy of the other protomer. This is the primary mechanism through which terpene pre-loading would operate—A2a agonism by terpenes altering CB1 signaling properties. Type II interactions are ligand-independent or constitutive, arising from the basal conformation of the heteromer, and are exemplified by the Kőfalvi finding that CB1's glutamate release inhibition depends on A2a's constitutive activity. Type III interactions operate through shared plasma membrane effectors such as G proteins and second messenger cascades. All three types would contribute to the pre-loading effect, but Type I interactions—the ligand-dependent

allosteric changes—are most directly manipulable through the timing and dose of terpene administration.

E. Structural Gaps and the A2a-CB2 Frontier

A critical limitation of the current heteromer literature is the absence of a high-resolution experimental structure. No cryo-EM or X-ray crystallographic structure of the A2a-CB1 heteromer exists as of early 2026; the Kőfalvi computational model, based on available A2a and CB1 homodimer structures, remains the best structural description. While the proximity ligation assay data (Moreno et al., 2018) unequivocally establish physical proximity between A2a and CB1 in native tissue, and the functional data demonstrate that co-expression produces signaling properties distinct from either receptor alone, the precise interface contacts, conformational dynamics, and allosteric transmission pathways remain computationally inferred rather than experimentally resolved.

The recently characterized A2a-CB2 heteromer (Llinas Del Torrent, 2024/2025; Rivas-Santisteban, 2025) opens an additional frontier relevant to the unified framework. β-Caryophyllene, the dietary cannabinoid with Ki = 155 ± 4 nM at CB2 (Gertsch et al., 2008; Section 1), could theoretically interact with A2a-CB2 heteromers in immune cells and peripheral tissues, providing a molecular substrate for the combined anti-inflammatory effects of terpene blends containing both A2a agonists (limonene) and CB2 agonists (β-caryophyllene). CBD has been shown to act as a biased allosteric modulator at this heteromer as well, suggesting that the heteromer framework extends beyond the CNS applications emphasized in this section.

F. The Heteromer as the Convergence Point

The A2a-CB1 heteromer system provides the missing molecular link between the parallel pharmacological pathways documented throughout this review. Terpenes activate A2a receptors (Schwarz et al., 2024; Park et al., 2011). Cannabinoids activate CB1 receptors. Where these

receptors form heteromers—striatopallidal neurons, hippocampal CA1 synapses—the two signaling streams converge and interact through defined allosteric mechanisms. Where heteromers are absent—striatonigral neurons, corticostriatal projections—the two pathways operate independently.

This architecture explains three otherwise puzzling observations. First, it explains why the entourage effect is selective rather than global: terpenes modulate only those CB1 effects mediated through heteromerized receptors. Second, it explains why CBD attenuates THC's cognitive effects (Aso et al., 2019) but not its motor or anxiolytic effects: hippocampal heteromers mediate cognition while other endpoints are mediated through non-heteromerized receptor populations. Third, it provides a molecular basis for the pre-loading paradigm: temporal separation of A2a and CB1 activation exploits the Type I allosteric properties of the heteromer to achieve selective modulation that would not occur with simultaneous exposure. The heteromer is not merely an interesting structural curiosity; it is the convergence architecture that makes terpene-cannabinoid crosstalk pharmacologically coherent.

Table 6. Evidence establishing the A2a-CB1 heteromer as the molecular substrate for terpene-cannabinoid crosstalk. PLA, proximity ligation assay; KO, knockout; AC, adenylyl cyclase; LTP, long-term potentiation; CA1, cornu ammonis 1 subfield of hippocampus.

Year	Study	Model / Approach	Key Finding	Significance for Unified Framework
2016	Chiodi et al. (J. Neurochem.)	Transgenic rats overexpressing A2a	CB1 agonist effects significantly reduced in A2a-overexpressing rats	A2a expression level modulates CB1 signaling efficiency

2017 Mouro et al. (Neuropharmacology) Chronic WIN55 + istradefylline, hippocampal LTP A2a antagonism (3 mg/kg, 28 days) reverted WIN-induced recognition memory deficits; SCH 58261 (100 nM) partially rescued CA1 LTP impairment A2a-CB1 interaction mediates cannabinoid memory effects; pharmacological rescue possible

2018 Moreno et al. (Neuropsychopharmacology) PLA, conditional CB1 KO mice, dorsal striatum A2a-CB1 heteromers present in striatopallidal (indirect) but absent from striatonigral (direct) and corticostriatal neurons; co-stimulation → cross-antagonism (↓↓ ERK/Akt); heteromer couples to Gq Definitive brain region mapping; pathway-specific distribution explains selective modulation

2018 Carlin et al. (Neuropharmacology) A2a agonism, peripheral receptors A2a agonism causes hypothermia via peripheral receptors Connects A2a to cannabinoid tetrad hypothermia component

2019 Aso et al. (Mol. Neurobiol.) Hippocampal CA1 neurons, CBD + THC A2a-CB1 heteromers at presynaptic level in CA1; CBD blunts THC cognitive impairment in A2a-dependent manner; CBD does NOT modify motor or anxiety—only cognition Brain region–dependent functional specificity: hippocampal = cognition, striatal = motor

2020 Kőfalvi et al. (BMC Biol.) Computational model, functional assays $(A2a)_2$-$(CB1)_2$ heterotetramer structure; A2a C-terminus determines constitutive AC activation via Gs; CB1 glutamate release inhibition depends on counteracting A2a's constitutive AC Quaternary structure established; CB1 function mechanistically dependent on A2a architecture

2023 Ferré et al. (Trends Pharmacol. Sci.) Theoretical framework Three allosteric interaction types formalized: Type I (ligand-dependent), Type II (constitutive), Type III (via membrane effectors) Provides framework for predicting pre-loading effects at the heteromer

2023 Raïch et al. (Int. J. Mol. Sci.) In vitro, ERK phosphorylation CBD (200 nM) decreased A2a-downstream ERK phosphorylation by ~40% Quantitative CBD–A2a modulation at nanomolar concentrations

2024 Sánchez-Fernández et al. (Acta Neuropsychiatr.) A2a allosteric modulation CBD acts as negative allosteric modulator of A2a: decreases agonist (CGS21680) potency 3.5–5× without acting as agonist itself Molecular mechanism for Aso finding: CBD modulates THC via A2a within heteromer ------ ---------- ---------- ---------- ---------- ---------- ---------- ----------

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Figure 4. The A2a-CB1 heteromer: architecture, brain distribution, and functional consequences. Panel A: Quaternary $(A2a)_2$-$(CB1)_2$ heterotetramer structure (Kőfalvi et al., 2020). A2a protomers couple exclusively to Gs (EC_{50} = 0.35 μM), CB1 protomers couple exclusively to Gi (EC_{50} = 1.05 μM), while the heteromer additionally engages Gq. A2a C-terminus determines constitutive adenylyl cyclase activation. Panel B: Brain region–specific heteromer distribution. Striatum: heteromers present in striatopallidal (indirect) but absent from striatonigral (direct) pathway neurons (Moreno et al., 2018). Hippocampus CA1: presynaptic heteromers mediate CBD's selective attenuation of THC cognitive impairment (Aso et al., 2019). CBD acts as negative allosteric modulator of A2a (Sánchez-Fernández et al., 2024). Panel C: Predicted pre-loading mechanism—terpene-mediated A2a activation 30 minutes before cannabis consumption establishes allosteric state that selectively modulates subsequent CB1 signaling. Three allosteric interaction types (Ferré et al., 2023) provide the theoretical framework. Panel D: Timeline of key evidence establishing the heteromer as the terpene-cannabinoid interface (2016–2024).

Section 6 word count: ~3,000 words | Cumulative (Sections 1–6): ~18,600 words

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Year	Study	Model / Approach	Key Finding	Significance for Unified Framework
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2017	Mouro et al. (Neuropharmacology)	Chronic WIN55 + istradefylline, hippocampal LTP	A2a antagonism (3 mg/kg, 28 days) reverted WIN-induced recognition memory deficits; SCH 58261 (100 nM) partially rescued CA1 LTP impairment	A2a-CB1 interaction mediates cannabinoid memory effects; pharmacological rescue possible
2018	Moreno et al. (Neuropsychopharmacology)	PLA, conditional CB1 KO mice, dorsal striatum	A2a-CB1 heteromers present in striatopallidal (indirect) but absent from striatonigral (direct) and corticostriatal neurons; co-stimulation → cross-antagonism (↓↓ ERK/Akt); heteromer couples to Gq	Definitive brain region mapping; pathway-specific distribution explains selective modulation

2018	Carlin et al. (Neuropharmacology)	A2a agonism, peripheral receptors	A2a agonism causes hypothermia via peripheral receptors	Connects A2a to cannabinoid tetrad hypothermia component
2019	Aso et al. (Mol. Neurobiol.)	Hippocampal CA1 neurons, CBD + THC	A2a-CB1 heteromers at presynaptic level in CA1; CBD blunts THC cognitive impairment in A2a-dependent manner; CBD does NOT modify motor or anxiety—only cognition	Brain region–dependent functional specificity: hippocampal = cognition, striatal = motor
2020	Kőfalvi et al. (BMC Biol.)	Computational model, functional assays	$(A2a)_2$-$(CB1)_2$ heterotetramer structure; A2a C-terminus determines constitutive AC activation via Gs; CB1 glutamate release inhibition depends on counteracting A2a's constitutive AC	Quaternary structure established; CB1 function mechanistically dependent on A2a architecture
2023	Ferré et al. (Trends Pharmacol. Sci.)	Theoretical framework	Three allosteric interaction types formalized: Type I (ligand-dependent), Type II (constitutive), Type III (via membrane effectors)	Provides framework for predicting pre-loading effects at the heteromer
2023	Raïch et al. (Int. J. Mol. Sci.)	In vitro, ERK phosphorylation	CBD (200 nM) decreased A2a-downstream ERK phosphorylation by ~40%	Quantitative CBD–A2a modulation at nanomolar concentrations
2024	Sánchez-Fernández et al. (Acta Neuropsychiatr.)	A2a allosteric modulation	CBD acts as negative allosteric modulator of A2a: decreases agonist (CGS21680) potency 3.5–5× without acting as agonist itself	Molecular mechanism for Aso finding: CBD modulates THC via A2a within heteromer

Section 7: Network Pharmacology

Pharmacological Reviews (ASPET)

Prepared for submission to Pharmacological Reviews (ASPET)

VII. Network Pharmacology Provides the Intellectual Framework for Cross-Botanical Terpene Design

A. From Single-Target to Multi-Target: The Network Pharmacology Revolution

The preceding sections have established that individual terpenes engage multiple receptor systems—β-caryophyllene at CB2, d-limonene at A2a, linalool at GABA-A, α-pinene at AChE and BDNF/TrkB, 1,8-cineole at TRPM8—and that these parallel pharmacological pathways converge at specific molecular junctions such as the A2a-CB1 heteromer. The question that naturally arises is whether this multi-target activity represents a limitation to be overcome or a design feature to be exploited. Network pharmacology provides the answer.

Hopkins (2008) introduced the concept of network pharmacology in a landmark paper that has accumulated over 3,310 citations. His central argument was that biological systems exhibit robustness—the ability to maintain function despite perturbation—precisely because they are organized as redundant, interconnected networks rather than linear pathways. This robustness makes single-target drugs inherently limited: blocking one node in a network merely shifts flux through alternative routes. The therapeutic implication is that modulating multiple network nodes simultaneously offers superior efficacy and reduced resistance compared to high-affinity single-target agents.

Wagner and Ulrich-Merzenich (2009), in a similarly influential contribution (over 1,495 citations), identified four specific mechanisms through which phytopharmaceutical synergy operates: multi-target effects on different components of the same disease pathway, pharmacokinetic enhancement of one compound by another, improvement of bioavailability

through facilitated absorption or reduced degradation, and circumvention of resistance mechanisms. Cannabis was explicitly cited as an illustrative example. All four mechanisms are operative in the terpene-cannabinoid system documented in this review: terpenes engage multiple receptor targets in the pain-anxiety-inflammation network (multi-target), they serve as penetration enhancers for co-inhaled compounds (pharmacokinetic), they achieve 60–65% pulmonary bioavailability through inhalation (bioavailability), and no terpene resistance mechanisms have been documented (resistance circumvention).

B. Terpenes as Natural Designed Multiple Ligands

Gertsch (2011) made the explicit connection between network pharmacology and terpenoid biochemistry, proposing that botanical drugs function as "intelligent mixtures" that are inherently network pharmacology agents. This framing inverts the conventional pharmaceutical perspective: rather than viewing terpenes' moderate affinity across multiple targets as poor selectivity—a deficiency compared to the high-affinity, single-target ideal of rational drug design—Gertsch argued that this distributed activity profile is an evolved feature that produces therapeutic effects through simultaneous modulation of multiple network nodes.

Anighoro et al. (2014) formalized this concept as polypharmacology, with Designed Multiple Ligands (DMLs) as an emerging pharmaceutical paradigm. DMLs are compounds intentionally designed to modulate multiple targets, in contrast to the traditional one-drug-one-target approach. Terpenes, with their moderate binding affinities across CB2 (Ki = 155 nM for β-caryophyllene), A2a, GABA-A, TRP channels, and serotonergic receptors, represent natural DMLs—molecules whose multi-target profile has been shaped by evolutionary pressures rather than medicinal chemistry optimization. The LaVigne et al. (2021) finding that individual terpenes produce the full cannabinoid tetrad—engaging analgesic, hypothermic, cataleptic, and locomotor endpoints simultaneously through different receptor mechanisms—is a textbook illustration of DML pharmacology in action.

The unified terpene pharmacology framework extends the DML concept in a direction not contemplated by Anighoro and colleagues: cross-botanical DML blending. If a single terpene is a natural DML engaging multiple targets, then a blend of terpenes from different botanical sources—essential oils combined with cannabis terpenes—constitutes a combinatorial DML library whose multi-target coverage can be rationally optimized. This is the pharmacological logic underlying strain-matched essential oil formulations: selecting botanical terpene sources that complement and extend the receptor coverage of a given cannabis chemotype's native terpene profile.

C. Cross-Plant Enhancement: Historical Precedent and Molecular Validation

The proposition that combining molecules from different botanical sources can produce pharmacological synergy is not merely theoretical. The piperine-curcumin paradigm provides the premier empirical precedent. Shoba et al. (1998) demonstrated that piperine from black pepper (Piper nigrum) increased curcumin bioavailability by 154% in rats and an extraordinary 2,000% in humans by inhibiting hepatic and intestinal glucuronidation—a pharmacokinetic enhancement mechanism of exactly the type Wagner and Ulrich-Merzenich subsequently categorized. This single study transformed curcumin from a compound with promising in vitro activity but negligible oral bioavailability into a viable pharmaceutical candidate, and it did so through a cross-botanical strategy: using a compound from one plant species to enhance the pharmacology of a compound from another.

Traditional Chinese Medicine provides millennia of empirical precedent through the Jun-Chen-Zuo-Shi combinatorial framework, which organizes multi-herb formulas into hierarchical roles: Jun (emperor/principal), Chen (minister/assistant), Zuo (adjuvant), and Shi (courier/guide). The Chinese Pharmacopoeia contains 1,444 such multi-herb formulas (Zhou et al., 2016), each representing a refined combination of compounds from different botanical sources optimized over centuries of clinical observation. Li et al. (2018) provided molecular validation through

systems pharmacology analysis of TCM formulas, confirming that different compounds from different source organisms share disease-relevant targets enriched in the same pathways—the molecular basis for cross-botanical synergy.

The terpene pre-loading paradigm described in this review follows the same cross-botanical logic: lavender-derived linalool priming GABA-A and serotonergic pathways before cannabis-derived linalool arrives through the same olfactory route, or black pepper-derived β-caryophyllene activating CB2 receptors before cannabis-derived β-caryophyllene reinforces the same signal. The molecules are identical regardless of source; the innovation lies in the temporal and quantitative strategy of supplementing cannabis terpene delivery with external botanical sources to overcome the bioavailability limitations documented in Section 2.

D. Mathematical Frameworks for Terpene Combination Optimization

Translating the network pharmacology concept into quantitative terpene blend design requires mathematical frameworks capable of distinguishing synergy from additivity from antagonism. Three complementary approaches are available, each suited to different aspects of the terpene combination problem.

Isobolographic analysis (Tallarida, 2006) uses dose equivalence to identify synergistic combinations: if the effective dose of two compounds administered together falls below the isobole—the line connecting their individual effective doses—the combination is synergistic. This approach is most appropriate when the combined agents share the same target, making it applicable to combinations of multiple A2a agonist terpenes (limonene plus geraniol, for example) or multiple GABA-A modulators (linalool plus α-pinene). The mathematical basis is Loewe Additivity, which assumes that compounds with the same mechanism of action will produce additive effects at their dose-equivalent ratios.

For terpenes acting through different receptor targets—the more common scenario in practice—the Bliss Independence model (1939) may be more pharmacologically appropriate.

Bliss Independence assumes that two agents with different mechanisms produce independent effects, and predicts that their combined effect equals the product of their individual effects. This model better captures the situation when, for example, a limonene-A2a interaction and a β-caryophyllene-CB2 interaction are operating through genuinely independent signaling cascades.

The Chou-Talalay Combination Index (Chou, 2006) provides a unified framework that subsumes both Loewe and Bliss models through the median-effect equation. The Combination Index (CI) classifies interactions quantitatively: CI < 1 indicates synergy, CI = 1 indicates additivity, and CI > 1 indicates antagonism. This framework, published in Pharmacological Reviews—the same journal to which this review is submitted—has been applied across oncology, virology, and antimicrobial pharmacology and is directly applicable to terpene combination optimization. Schwarz et al.'s (2024) demonstration that low-dose terpene (100 mg/kg) combined with low-dose morphine (3.2 mg/kg) produced synergistic antinociception is precisely the type of combination interaction that isobolographic analysis and the Chou-Talalay framework are designed to quantify.

Cheng et al. (2019) introduced a computational approach using human protein-protein interaction networks (the interactome) to predict effective drug combinations. Their key finding was that compounds whose targets are proximal in the interactome—connected by short network paths—are more likely to produce synergistic combinations. This approach is directly applicable to terpene blend optimization: mapping the interactome proximity between A2a (limonene target), CB1 and CB2 (β-caryophyllene targets), GABA-A (linalool target), and TRP channels (1,8-cineole target) would predict which terpene combinations are most likely to synergize and which might antagonize. No published study has yet applied interactome-based combination prediction to terpene pharmacology, representing an immediate opportunity for computational analysis.

E. Applying Network Pharmacology to the Unified Framework

The network pharmacology perspective reframes every aspect of the evidence reviewed in Sections 1 through 6. The multi-receptor terpene landscape of Section 1 is not a catalog of scattered activities but a polypharmacological profile amenable to rational optimization. The inhalation pharmacokinetics of Section 2 describe the bioavailability parameters of a multi-component delivery system. The aromatherapy RCTs of Section 3 and forest bathing immunology of Section 5 document the clinical effects of natural terpene mixtures—essential oils and forest air being inherently multi-component exposures—through the lens of network-level pharmacology. The entourage effect of Section 4 is network pharmacology operating within a single botanical source, while the A2a-CB1 heteromer of Section 6 provides the molecular junction where two major network nodes physically converge.

The practical application takes two forms. First, within-strain optimization: given a cannabis chemotype's terpene profile from laboratory analysis, network pharmacology principles predict which receptor systems are engaged and at what relative efficacy, enabling design of supplementary essential oil blends that reinforce the dominant pathway while adding coverage of under-represented targets. A limonene-dominant strain would primarily engage A2a; a supplementary blend enriched in linalool (GABA-A) and β-caryophyllene (CB2) would extend the receptor coverage into anxiolytic and anti-inflammatory domains. Second, cross-strain comparison: network pharmacology provides a quantitative framework for comparing chemotypes not by their traditional indica/sativa classification, which Jikomes et al. (2022) demonstrated is chemically meaningless across 89,923 commercial samples, but by their multi-target receptor engagement profiles—a pharmacologically grounded taxonomy that connects cannabis science directly to the receptor pharmacology established in the aromatherapy and forest medicine literatures.

The absence of any published application of formal network pharmacology methods—interactome analysis, synergy quantification, or DML optimization—to the cross-botanical

terpene framework underscores that this field remains in its earliest conceptual stages. The mathematical tools exist, the receptor data are accumulating, and the clinical precedents from both traditional medicine and modern phytopharmacology are well established. What remains is the systematic application of these frameworks to the specific terpene-receptor-pathway data compiled in this review.

Table 7. Network pharmacology frameworks applicable to cross-botanical terpene design. The foundational concepts (Hopkins, Wagner, Gertsch) provide the theoretical justification for multi-target terpene pharmacology. Cross-plant precedents (piperine-curcumin, TCM) establish that cross-botanical enhancement is pharmacologically valid. Mathematical frameworks (Tallarida, Bliss/Chou, Cheng) provide the quantitative tools for optimizing terpene blend compositions. DML, Designed Multiple Ligand; TCM, Traditional Chinese Medicine; CI, Combination Index.

--- --- --- --- Source Concept / Framework Core Principle Application to Unified Terpene Pharmacology

Hopkins 2008 (Nat. Chem. Biol.) Network pharmacology coined; >3,310 citations Biological robustness makes single-target drugs inherently limited; multi-node modulation offers superior efficacy Provides theoretical foundation: terpenes acting on A2a, CB1/CB2, GABA-A, TRP channels, 5-HT simultaneously are inherent network pharmacology agents

Wagner & Ulrich-Merzenich 2009 (Phytomedicine) Four synergistic mechanisms; >1,495 citations Multi-target effects, pharmacokinetic enhancement, bioavailability improvement, resistance circumvention; cannabis cited as illustration All four mechanisms

operative in terpene-cannabinoid system: multi-receptor engagement, penetration enhancement, pulmonary bioavailability, and no resistance documented

Gertsch 2011 (Planta Med.) Terpenoid network pharmacology Botanical drugs as “intelligent mixtures” functioning as inherent network pharmacology agents; terpenoids specifically proposed Explicit bridge: terpenoid polypharmacology is not a limitation to overcome but an evolved design feature to exploit

Anighoro et al. 2014 (J. Med. Chem.) Designed Multiple Ligands (DMLs) Polypharmacology formalized; DMLs modulate multiple targets by design rather than accident Terpenes with moderate affinity across CB2, A2a, GABA-A, TRP channels represent natural DMLs; cross-botanical blending extends the DML concept across source organisms

Shoba et al. 1998 (Planta Med.) Piperine-curcumin paradigm Piperine from black pepper increased curcumin bioavailability 154% in rats, 2,000% in humans by inhibiting hepatic/intestinal glucuronidation Premier precedent for cross-botanical enhancement; demonstrates that compounds from different plants can synergistically enhance each other’s pharmacology

Zhou et al. 2016 (Front. Pharmacol.) TCM Jun-Chen-Zuo-Shi framework 1,444 multi-herb formulas in Chinese Pharmacopoeia; millennia of empirical optimization of cross-plant combinations Historical precedent: traditional medicine systems have long combined molecules from different botanical sources for therapeutic synergy

Li et al. 2018 (Sci. Rep.) Systems pharmacology of TCM Different compounds from different source organisms share disease-relevant targets enriched in the same pathways Molecular validation of cross-botanical synergy principle: shared targets regardless of botanical origin

Tallarida 2006 (JPET) Isobolographic analysis Dose equivalence method to identify synergy (below isobole) vs. antagonism (above isobole) Applicable to same-target terpene combinations (e.g., multiple A2a agonists); Loewe Additivity model

Bliss 1939; Chou 2006 (Pharmacol. Rev.) Independence / Combination Index Bliss Independence for different-target agents; Chou-Talalay CI provides unified framework using median-effect equation Different-target model more appropriate for terpene blends acting on A2a + CB2 + GABA-A simultaneously; CI <1 = synergy, =1 additive, >1 antagonism

Cheng et al. 2019 (Nat. Commun.) Network-based drug combination prediction Human interactome proximity predicts effective drug combinations; closer network distance = higher combination efficacy Directly applicable to terpene blend optimization: target proximity in protein-protein interaction networks predicts which terpene combinations will synergize ---

Section 7 word count: ~2,800 words | Cumulative (Sections 1–7): ~20,400 words

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Source	Concept / Framework	Core Principle	Application to Unified Terpene Pharmacology

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Wagner & Ulrich-Merzenich 2009 (Phytomedicine)	Four synergistic mechanisms; >1,495 citations	Multi-target effects, pharmacokinetic enhancement, bioavailability improvement, resistance circumvention; cannabis cited as illustration	All four mechanisms operative in terpene-cannabinoid system: multi-receptor engagement, penetration enhancement, pulmonary bioavailability, and no resistance documented
Gertsch 2011 (Planta Med.)	Terpenoid network pharmacology	Botanical drugs as “intelligent mixtures” functioning as inherent network pharmacology agents; terpenoids specifically proposed	Explicit bridge: terpenoid polypharmacology is not a limitation to overcome but an evolved design feature to exploit
Anighoro et al. 2014 (J. Med. Chem.)	Designed Multiple Ligands (DMLs)	Polypharmacology formalized; DMLs modulate multiple targets by design rather than accident	Terpenes with moderate affinity across CB2, A2a, GABA-A, TRP channels represent natural DMLs; cross-botanical blending extends the DML concept across source organisms
Shoba et al. 1998 (Planta Med.)	Piperine-curcumin paradigm	Piperine from black pepper increased curcumin bioavailability 154% in rats, 2,000% in humans by inhibiting hepatic/intestinal glucuronidation	Premier precedent for cross-botanical enhancement; demonstrates that compounds from different plants can synergistically enhance each other’s pharmacology
Zhou et al. 2016 (Front. Pharmacol.)	TCM Jun-Chen-Zuo-Shi framework	1,444 multi-herb formulas in Chinese Pharmacopoeia; millennia of empirical optimization of cross-plant combinations	Historical precedent: traditional medicine systems have long combined molecules from different botanical sources for therapeutic synergy

Li et al. 2018 (Sci. Rep.)	Systems pharmacology of TCM	Different compounds from different source organisms share disease-relevant targets enriched in the same pathways	Molecular validation of cross-botanical synergy principle: shared targets regardless of botanical origin
Tallarida 2006 (JPET)	Isobolographic analysis	Dose equivalence method to identify synergy (below isobole) vs. antagonism (above isobole)	Applicable to same-target terpene combinations (e.g., multiple A2a agonists); Loewe Additivity model
Bliss 1939; Chou 2006 (Pharmacol. Rev.)	Independence / Combination Index	Bliss Independence for different-target agents; Chou-Talalay CI provides unified framework using median-effect equation	Different-target model more appropriate for terpene blends acting on A2a + CB2 + GABA-A simultaneously; CI <1 = synergy, =1 additive, >1 antagonism
Cheng et al. 2019 (Nat. Commun.)	Network-based drug combination prediction	Human interactome proximity predicts effective drug combinations; closer network distance = higher combination efficacy	Directly applicable to terpene blend optimization: target proximity in protein-protein interaction networks predicts which terpene combinations will synergize

Section 8: Funding Landscape

Pharmacological Reviews (ASPET)

Prepared for submission to Pharmacological Reviews (ASPET)

VIII. Funding Landscape Faces Existential Threat Despite Emerging Translational Momentum

A. The NCCIH Foundation and Its Proposed Elimination

The federal infrastructure supporting terpene pharmacology research traces primarily to a single NIH institute: the National Center for Complementary and Integrative Health (NCCIH). The NCCIH Notice of Special Interest NOT-AT-22-027, now expired, represented the first dedicated federal funding solicitation for cannabis terpene research. The initial request for applications round spawned approximately $3 million in grants across nine awards, and listed ten specific terpenes as priorities for investigation: myrcene, β-caryophyllene, limonene, α-terpineol, linalool, α-phellandrene, α-pinene, β-pinene, γ-terpinene, and α-humulene. Key funded investigators include Sara Jane Ward at Temple University, Jenny Wiley at RTI International, Cassandra Quave at Emory University in collaboration with Isaac Chiu at Harvard, John Streicher at the University of Arizona, Ziva Cooper at UCLA, and David Sarlah at the University of Illinois Urbana-Champaign.

The most consequential grant in the current terpene pharmacology landscape is Streicher's R01AT011517, funded by NCCIH, which remains active and has supported the LaVigne, Schwarz, and Seekins progression that discovered the adenosine A2a receptor mechanism documented in Sections 4 and 6 of this review. This single grant's productivity illustrates both the potential of targeted terpene funding and the vulnerability of concentrating the field's most important mechanistic research in a single investigator supported by a single institute. Streicher's disclosed conflicts—consultation for Napreva, a terpene product company,

and equity positions in Botanical Results and Teleport Pharmaceuticals—represent the emerging translational ecosystem around terpene pharmacology but also warrant the standard scrutiny applied to industry-academic relationships in any field.

The R3CR (Resource Center for Cannabis and Cannabinoid Research) launched in April 2025 under grant U24AT013161, co-funded by NCCIH with contributions from NCI, NIA, and NIDA. Hosted at the University of Mississippi with partnerships at Washington State University and the United States Pharmacopeia, the R3CR provides three operational cores: Regulatory Guidance to help investigators navigate the complex federal and state requirements for cannabis research, Research Support including seed funding for pilot studies, and Research Standards for analytical chemistry and product characterization. The first seed funding proposals were due December 8, 2025, meaning the center had been accepting applications for less than eight months before its primary sponsor faced proposed elimination.

That proposed elimination represents the existential threat to the field. The FY2026 White House budget proposes the complete elimination of NCCIH—the primary NIH institute funding terpene research. By mid-2025, approximately 2,300 NIH grants representing over $3.8 billion had been terminated or frozen across all institutes. While Congress has pushed back on the NCCIH elimination, the outcome remains uncertain as of February 2026. The implications for terpene pharmacology are severe: NCCIH funded the A2a mechanism discovery, the R3CR infrastructure, and the majority of the investigator-initiated grants that comprise the field's pipeline. Loss of this institute would eliminate the primary federal mechanism for terpene research funding with no clear alternative sponsor, as NIDA's mission centers on substance use disorders rather than the broader pharmacological characterization of plant secondary metabolites.

The NIDA cancellation of the 57-year University of Mississippi cannabis production contract further destabilizes the research infrastructure. Since 1968, the University of Mississippi had served as the sole federally authorized source of research-grade cannabis, and while the DEA began licensing additional cultivators in 2021, the loss of this established program disrupts

the supply chain for standardized research materials at a time when the field is transitioning from preclinical to clinical investigation.

B. The Johns Hopkins Behavioral Pharmacology Research Unit

The Cannabis Science Laboratory at Johns Hopkins University, led by Ryan Vandrey and Tory Spindle within the Behavioral Pharmacology Research Unit (BPRU), operates under NIDA P50DA056408 and R01DA043475. This laboratory has produced the field's most important human data: the Spindle et al. (2024) demonstration of selective limonene-mediated anxiolysis during THC administration (Section 4) and the Kumar et al. (2025) negative result showing α-pinene did not attenuate THC-induced memory impairment. Jenkins et al. (2023) tested five terpenes preclinically within the same program, finding that bisabolol and d-limonene showed anxiolytic effects—data that informed the subsequent human trial design.

The BPRU pipeline extends in two directions. Planned studies include oral d-limonene replication, testing whether the anxiolytic effect observed with vaporized administration translates to oral delivery, and myrcene sedation investigation, addressing the widespread but largely anecdotal claim that myrcene-dominant cannabis strains produce sedation. A patent application (PCT/US2022/014296) has been filed for the use of d-limonene to reduce THC-induced anxiety, representing the most direct translational step from the academic terpene literature toward commercial application. The filing of this patent underscores the commercial significance of the Spindle finding: if limonene can reliably reduce cannabis-associated anxiety at the 15 mg dose level, this has immediate implications for cannabis product formulation, ancillary product development, and clinical anxiety management.

The Hopkins program's funding through NIDA rather than NCCIH provides partial insulation from the proposed NCCIH elimination, though the broader NIH funding environment remains precarious. The IND pathway has been successfully navigated for terpene administration

in humans (IND #140339), establishing a regulatory precedent that reduces barriers for subsequent investigators seeking to conduct human terpene-cannabinoid interaction studies.

C. State Funding: The California Model

State-level cannabis tax revenue has created an alternative funding mechanism that partially compensates for federal constraints. The California Department of Cannabis Control (DCC) has invested nearly $80 million since 2020 across three grant rounds. The 2025 round awarded approximately $30 million for 22 projects, several of which include terpene-relevant components. UCLA received $1.23 million for a Flower-Flavor-Compound Reference Dataset that will systematically catalog terpene profiles across commercial cannabis cultivars—precisely the type of chemical characterization data needed to connect laboratory pharmacology with real-world product variability. San Diego State University received $1.16 million for a cannabis worker terpene exposure assessment, providing occupational health data on chronic inhalation of terpenes at workplace concentrations.

The California model demonstrates that state cannabis tax revenue can sustain meaningful research investment, but it carries inherent limitations. State funding is restricted to state-relevant questions, typically prioritizing public health and regulatory compliance over basic mechanistic pharmacology. The DCC portfolio emphasizes product safety, agricultural practices, and social equity—all important priorities, but none capable of funding the receptor pharmacology, heteromer characterization, or controlled human trials that have produced the field’s most significant advances. State funding complements but cannot substitute for the NIH infrastructure that supports investigator-initiated mechanistic research.

D. The Regulatory Advantage: GRAS Status and the Scheduling Paradox

A distinctive feature of terpene pharmacology—one that distinguishes it from virtually every other area of cannabis-adjacent research—is the regulatory freedom afforded by GRAS

(Generally Recognized as Safe) status. Most individual terpenes are classified as GRAS under FDA 21 CFR 172.510 and FEMA evaluations. β-Caryophyllene received explicit GRAS status, and hemp-derived terpenes are unregulated under the 2018 Farm Bill. This means that the molecules whose pharmacology is documented in this review can be freely purchased, formulated, and administered without DEA registration, FDA IND application, or Schedule I authorization—provided they are sourced from non-cannabis botanicals.

The paradox is stark: identical molecules are GRAS food additives and unrestricted essential oil ingredients when derived from lavender, black pepper, or citrus, yet become Schedule I controlled substances when present in cannabis containing more than 0.3% THC. This regulatory asymmetry creates both an obstacle and an opportunity. The obstacle is that cannabis terpene research still requires the full Schedule I authorization apparatus (DEA registration, FDA IND, state approvals, and until recently, sole-source material from the University of Mississippi). The opportunity is that the cross-botanical terpene framework proposed in this review—using essential oil–derived terpenes to supplement cannabis terpene exposure—operates entirely within the GRAS regulatory space. Essential oil blends designed to complement cannabis terpene profiles require no controlled substance authorization to manufacture, sell, or study, even though they contain the same pharmacologically active molecules.

On December 18, 2025, Executive Order 14370 directed the Attorney General to expedite cannabis reclassification from Schedule I to Schedule III. If implemented, rescheduling would dramatically reduce the regulatory burden on cannabis terpene research by eliminating the DEA registration requirement and potentially expanding the sources of research-grade cannabis. However, the rulemaking timeline remains uncertain as of February 2026, and rescheduling to Schedule III would not eliminate all regulatory barriers—cannabis would still require prescription for medical use and would remain subject to DEA manufacturing quotas. The terpenes themselves, notably, would remain GRAS regardless of their botanical source, meaning

the regulatory advantage of the cross-botanical approach persists even in a post-rescheduling environment.

E. The Funding Gap: What Cannot Be Supported

The current funding landscape reveals a systematic gap between the questions the field most urgently needs to answer and the mechanisms available to support that research. Three categories of essential investigation lack adequate funding infrastructure.

First, cross-disciplinary integration studies. No current federal funding mechanism is designed to support research that spans aromatherapy, cannabis pharmacology, and forest medicine. The NCCIH portfolio addresses complementary and integrative health but does not specifically solicit studies combining cannabis and non-cannabis terpene sources. NIDA focuses on substance use but not on the broader phytopharmacology of plant secondary metabolites. The cross-botanical terpene framework proposed in this review falls between these missions, potentially orphaned by the very disciplinary boundaries it seeks to dissolve.

Second, terpene pre-loading clinical trials. Testing the hypothesis that aromatherapy terpene pre-loading modulates subsequent cannabis effects requires a study design that combines essential oil administration (unregulated) with cannabis administration (Schedule I). The regulatory complexity of this hybrid protocol—requiring both FDA IND authorization for the cannabis component and standard human subjects review for the aromatherapy component—has no established precedent. The Hopkins IND #140339 covers terpene administration but not the combined essential oil–plus–cannabis protocol that would directly test the pre-loading paradigm.

Third, inhalation route pharmacology. The Schwarz et al. (2024) finding that terpene antinociception was not produced by the inhalation route in the mouse model represents the field’s most significant translational limitation, yet no funded study specifically addresses inhalation pharmacokinetics of terpene-cannabinoid combinations in humans. The Moss research program at Northumbria (Section 2) has demonstrated cognitive effects of inhaled terpenes in

over 400 participants but has not connected this work to the cannabis pharmacology literature, again reflecting the disciplinary boundaries that inhibit integrated research.

The convergence of proposed NCCIH elimination, NIDA contract cancellation, and uncertain cannabis rescheduling timeline creates a window of maximum vulnerability for terpene pharmacology research. The field has achieved unprecedented mechanistic clarity—the A2a receptor pathway, the heteromer architecture, the first controlled human evidence—precisely at the moment when the funding infrastructure that produced these advances faces existential threat. Whether this momentum survives the current funding environment will determine whether the unified terpene pharmacology framework proposed in this review remains a theoretical construct or becomes an experimentally validated paradigm.

Table 8. Current terpene research funding landscape and regulatory developments (as of February 2026). The field's primary federal sponsor (NCCIH) faces proposed elimination in the FY2026 budget, threatening the infrastructure that produced the A2a mechanism discovery and first controlled human terpene-cannabinoid interaction data. State funding through California DCC and regulatory advantages through GRAS status provide partial but insufficient alternatives. NCCIH, National Center for Complementary and Integrative Health; DCC, Department of Cannabis Control; BPRU, Behavioral Pharmacology Research Unit; GRAS, Generally Recognized as Safe; IND, Investigational New Drug; EO, Executive Order; AG, Attorney General.

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--- Mechanism / Grant Source Funding Level Key Investigators / Institutions Status and Significance

NCCIH NOT-AT-22-027 NIH/NCCIH ~$3M+ (9 awards); listed 10 terpenes Ward (Temple), Wiley (RTI), Quave/Chiu (Emory/Harvard), Streicher (Arizona), Cooper (UCLA), Sarlah (UIUC) EXPIRED; spawned foundational terpene pharmacology grants; NCCIH elimination threatens continuation

R01AT011517 NIH/NCCIH Active R01 (Streicher lab) John Streicher, University of Arizona Funded LaVigne → Schwarz → Seekins progression discovering A2a mechanism; PI consults for Napreva, holds equity in Botanical Results and Teleport Pharmaceuticals

R3CR (U24AT013161) NCCIH + NCI/NIA/NIDA New center grant, April 2025 University of Mississippi + WSU + USP Three cores: Regulatory Guidance, Research Support (seed funding), Research Standards; first proposals due Dec 8, 2025; vulnerable to NCCIH elimination

P50DA056408 NIDA P50 Center Grant Johns Hopkins BPRU (Vandrey, Spindle) Supports Cannabis Science Laboratory; Spindle 2024 limonene study; Kumar 2025 α-pinene study

R01DA043475 NIDA R01 Grant Johns Hopkins (Vandrey) Additional BPRU funding; planned oral d-limonene replication and myrcene sedation investigation

PCT/US2022/014296 Patent application N/A Johns Hopkins d-Limonene to reduce THC-induced anxiety; demonstrates translational momentum

California DCC Rounds 1–3 State (cannabis tax revenue) ~$80M total since 2020; 2025 round: ~$30M for 22 projects Multiple California institutions Terpene-relevant: UCLA Flower-Flavor-Compound Reference Dataset ($1.23M); SDSU cannabis worker terpene exposure ($1.16M)

IND #140339 FDA N/A Johns Hopkins terpene studies Successfully

navigated FDA IND pathway for terpene administration in humans; establishes regulatory precedent

FY2026 budget proposal White House Proposes complete NCCIH elimination N/A

EXISTENTIAL THREAT: NCCIH is primary NIH funder of terpene research; ~2,300 NIH grants ($3.8B+) terminated/frozen by mid-2025; outcome uncertain

EO 14370 (Dec 18, 2025) Executive branch N/A N/A Directs AG to expedite cannabis reclassification to Schedule III; rulemaking timeline uncertain as of Feb 2026

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Section 8 word count: ~2,700 words | Cumulative (Sections 1–8): ~22,400 words

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Mechanism / Grant	Source	Funding Level	Key Investigators / Institutions	Status and Significance

NCCIH NOT-AT-22-027	NIH/NCCIH	~$3M+ (9 awards); listed 10 terpenes	Ward (Temple), Wiley (RTI), Quave/Chiu (Emory/Harvard), Streicher (Arizona), Cooper (UCLA), Sarlah (UIUC)	EXPIRED; spawned foundational terpene pharmacology grants; NCCIH elimination threatens continuation
R01AT011517	NIH/NCCIH	Active R01 (Streicher lab)	John Streicher, University of Arizona	Funded LaVigne → Schwarz → Seekins progression discovering A2a mechanism; PI consults for Napreva, holds equity in Botanical Results and Teleport Pharmaceuticals
R3CR (U24AT013161)	NCCIH + NCI/NIA/NIDA	New center grant, April 2025	University of Mississippi + WSU + USP	Three cores: Regulatory Guidance, Research Support (seed funding), Research Standards; first proposals due Dec 8, 2025; vulnerable to NCCIH elimination
P50DA056408	NIDA	P50 Center Grant	Johns Hopkins BPRU (Vandrey, Spindle)	Supports Cannabis Science Laboratory; Spindle 2024 limonene study; Kumar 2025 α-pinene study

R01DA043475	NIDA	R01 Grant	Johns Hopkins (Vandrey)	Additional BPRU funding; planned oral d-limonene replication and myrcene sedation investigation
PCT/ US2022/014296	Patent application	N/A	Johns Hopkins	d-Limonene to reduce THC-induced anxiety; demonstrates translational momentum
California DCC Rounds 1–3	State (cannabis tax revenue)	~$80M total since 2020; 2025 round: ~$30M for 22 projects	Multiple California institutions	Terpene-relevant: UCLA Flower-Flavor-Compound Reference Dataset ($1.23M); SDSU cannabis worker terpene exposure ($1.16M)
IND #140339	FDA	N/A	Johns Hopkins terpene studies	Successfully navigated FDA IND pathway for terpene administration in humans; establishes regulatory precedent
FY2026 budget proposal	White House	Proposes complete NCCIH elimination	N/A	EXISTENTIAL THREAT: NCCIH is primary NIH funder of terpene research; ~2,300 NIH grants ($3.8B+) terminated/frozen by mid-2025; outcome uncertain

EO 14370 (Dec 18, 2025)	Executive branch	N/A	N/A	Directs AG to expedite cannabis reclassification to Schedule III; rulemaking timeline uncertain as of Feb 2026

Section 9: The 1937 Coincidence + Conclusion

Pharmacological Reviews (ASPET)

Prepared for submission to Pharmacological Reviews (ASPET)

IX. The 1937 Coincidence: How Regulatory History Created Three Parallel Disciplines Studying Identical Molecules

A. The Year of Divergence

In a remarkable historical coincidence, 1937 saw both the passage of the Marihuana Tax Act in the United States and the publication of René-Maurice Gattefossé's Aromathérapie in France. These two events, occurring independently on opposite sides of the Atlantic, established the regulatory and intellectual architecture that would fragment terpene science for nearly nine decades.

The Marihuana Tax Act, championed by Federal Bureau of Narcotics Director Harry J. Anslinger, imposed prohibitive taxes and bureaucratic barriers on cannabis that effectively ended most cannabis research in the United States. Cannabis was subsequently removed from the United States Pharmacopoeia in 1942. The regulatory apparatus that followed—culminating in cannabis's classification as Schedule I under the Controlled Substances Act of 1970—required DEA registration, sole-source material from the University of Mississippi (the only federally authorized cultivation facility until 2021), FDA Investigational New Drug application, and state-level approvals for any research involving cannabis-derived compounds, including the terpenes that constitute the plant's essential oil fraction.

Meanwhile, Gattefossé's book formalized the scientific study of these same molecules—linalool, pinene, limonene, and dozens of other terpenes—under the entirely separate banner of aromatherapy. Because these compounds were derived from lavender, pine, citrus, and other non-cannabis botanical sources, they required no special regulatory authorization. The result was

that pharmacological investigation of identical molecules proceeded along two entirely separate tracks: cannabis terpene research, confined by Schedule I restrictions to a small number of authorized laboratories, and aromatherapy research, freely conducted in university settings worldwide without DEA oversight.

A third discipline emerged in 1982, when Tomohide Akiyama of the Japanese Ministry of Agriculture, Forestry and Fisheries coined the term shinrin-yoku (forest bathing) to describe the health practice of immersion in forest environments. Formal research began in 2004 with Qing Li's first studies at Nippon Medical School. This third literature studied the same α-pinene and limonene molecules under yet another nomenclature—"phytoncides"—in yet another set of journals (Environmental Health and Preventive Medicine, International Journal of Environmental Research and Public Health), with yet another set of conferences and funding mechanisms. The disciplinary isolation is reflected in the nomenclature itself: Qing Li frames α-pinene and limonene as phytoncides, Ethan Russo describes them as cannabis terpenes, and Mark Moss calls them essential oil constituents. These are labels reflecting disciplinary boundaries, not chemical distinctions.

B. The Regulatory Architecture of Isolation

The regulatory asymmetry between cannabis-derived and non-cannabis-derived terpenes created the structural conditions for bibliometric isolation. Cannabis terpene research published in Cannabis and Cannabinoid Research, the Journal of Cannabis Research, and drug policy–oriented journals. Aromatherapy research appeared in the International Journal of Aromatherapy, Complementary Therapies in Medicine, and Phytomedicine. Forest medicine research occupied Environmental Health and Preventive Medicine and the International Journal of Environmental Research and Public Health. No published bibliometric study quantifying cross-citation rates between these three literatures was found during the preparation of this review, but the structural

isolation is evident from even casual examination of the reference lists in each field's major publications.

The practical consequence for pharmacology was that receptor-level findings in one field remained unknown to investigators in the others. Park et al.'s (2011) identification of limonene as an adenosine A2a agonist appeared in the botanical/food chemistry literature and was not connected to cannabis pharmacology until LaVigne et al. (2021) identified A2a involvement in terpene-mediated catalepsy a full decade later. Moss's extensive program demonstrating cognitive effects of inhaled 1,8-cineole in over 400 human participants (2003–2025) has not been cited in any cannabis terpene pharmacology paper, despite its direct relevance to the inhalation pharmacokinetics discussed in Section 2. Li's in vitro demonstration that α-pinene, 1,8-cineole, and d-limonene dose-dependently activate NK cells (2006) addresses immune modulation by the same terpenes studied in the cannabis entourage effect literature, yet the two bodies of evidence have never been integrated in a single analysis prior to this review.

C. Market Forces Reinforced the Divide

Commercial dynamics compounded the regulatory fragmentation. The essential oil industry grew to $26.91 billion globally by 2024 (Polaris Market Research), led by companies such as dōTERRA (founded 2008, approximately 5 million customers) and Young Living (founded 1993). These companies freely commercialized limonene, linalool, and α-pinene—the same molecules that cannabis researchers needed Schedule I authorization to study. The paradox is stark: identical molecules are GRAS food additives and unrestricted essential oil ingredients when derived from lavender or citrus, yet become Schedule I controlled substances when present in cannabis containing more than 0.3% THC.

The essential oil industry had strong commercial incentives to distance itself from cannabis associations. The wellness-oriented branding of aromatherapy products—sold through direct marketing networks, health food stores, and spa channels—depended on mainstream

consumer acceptance that any association with cannabis could jeopardize. Conversely, the cannabis industry initially had little interest in the aromatherapy literature, focused as it was on THC potency as the primary value proposition. The result was two parallel commercial ecosystems selling the same terpene molecules to different consumer populations through different distribution channels, each with economic incentives to maintain the separation.

D. Convergence Signals: 2019–2025

Multiple developments between 2019 and 2025 signal that the disciplinary boundaries are beginning to dissolve, though the process remains incomplete. The Leafly cannabis guide, the most influential cannabis consumer education platform in North America, abandoned the indica/sativa/hybrid classification system in approximately 2019–2020 in favor of terpene-based classification using eight color-coded terpene profiles. Their Flower Finder tool allows consumers to search by terpene preference rather than strain name—an implicit recognition that terpene profiles, not cultivar labels, determine experiential effects.

The Emerald Cup, California's most prestigious cannabis competition, restructured its 2021 judging categories around six terpene-based genres (myrcene-dominant, β-caryophyllene/limonene, limonene-dominant, myrcene/ocimene, terpinolene-dominant, and exotic profiles) in partnership with SC Labs. This represented an explicit rejection of the indica/sativa taxonomy at the premium end of the market. Jikomes et al. (2022) provided the empirical validation: their analysis of 89,923 commercial cannabis samples in PLOS ONE confirmed that terpene profiles—not indica/sativa labels—are the key chemical differentiators among commercial cannabis cultivars.

At the scientific level, the LaVigne–Schwarz–Spindle research trajectory documented in Section 4 has organically begun bridging the cannabis and non-cannabis terpene literatures. Schwarz et al.'s (2024) identification of adenosine A2a as the primary terpene analgesic mechanism connects directly to Park et al.'s (2011) earlier identification of limonene as an A2a

agonist in the non-cannabis literature. Schwarz et al. (2025) extended this by demonstrating that natural cannabis terpene blends produce antinociception through A2a—the first study to explicitly test cannabis-specific terpene combinations through the A2a mechanism identified in the aromatherapy-adjacent literature.

The psychedelic research renaissance provides an instructive parallel. After approximately thirty years of near-total dormancy following prohibition-era restrictions, psychedelic pharmacology experienced explosive growth: 54% of the top 100 most-cited psychedelic research papers were published between 2010 and 2020. However, the terpene fragmentation differs from the psychedelic case in a critical way. Psychedelic research was primarily temporal—the same field was interrupted and then resumed. Terpene fragmentation is primarily disciplinary—three parallel fields have been studying the same molecules continuously, but in isolation from each other. This makes reunification conceptually harder, because it requires not just resuming interrupted work but integrating independently developed bodies of knowledge with different methodological traditions, different nomenclatures, and different institutional structures. It also makes it potentially more impactful, because the convergent evidence base, once assembled, is far larger than any single field's contribution.

Table 9. Key events in disciplinary fragmentation and convergence, 1937–2024. The table traces the historical forces that created three parallel literatures studying identical terpene molecules, the market dynamics that reinforced the divide, and the recent convergence signals that make reunification both possible and urgent. MAFF, Ministry of Agriculture, Forestry and Fisheries; FBN, Federal Bureau of Narcotics.

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---------- Year Event

Description

Significance for Disciplinary Fragmentation / Convergence

1937 Marihuana Tax Act (US) FBN Director Anslinger imposed prohibitive taxes/bureaucratic barriers on cannabis; cannabis removed from US Pharmacopoeia 1942 Initiated regulatory isolation of cannabis terpene research; Schedule I authorization required for identical molecules freely available from other botanical sources

1937 Gattefossé's Aromathérapie published Formalized scientific study of linalool, pinene, limonene under the banner of aromatherapy; separate from cannabis pharmacology from inception Created parallel discipline studying identical terpene molecules without any connection to cannabis literature; separate journals, conferences, funding streams

1982 Shinrin-yoku coined (Akiyama, Japanese MAFF) Third discipline established; formal forest bathing research began with Li 2004
Added third parallel literature studying α-pinene and limonene as "phytoncides"—same molecules, third nomenclature, third set of journals

1993–2008 Essential oil industry growth (Young Living 1993, dōTERRA 2008) Industry grew to $26.91B globally by 2024 (Polaris Market Research); dōTERRA alone: 5 million customers Market forces commercialized terpenes outside cannabis regulatory framework; reinforced disciplinary divide by creating economic incentives for separation

2019–2020 Leafly abandons indica/sativa taxonomy Cannabis guide restructured around cannabinoid ratios and 8 color-coded terpene profiles; Flower Finder tool searches by terpene preference Most influential cannabis platform signals terpene primacy; implicit recognition that terpene profiles, not cultivar labels, determine effects

2021 Emerald Cup restructures judging by terpene Six terpene-based genres (myrcene, β-caryophyllene, limonene, terpinolene, α-pinene, β-ocimene) replace

indica/sativa; SC Labs partnership Premium cannabis competition validates terpene-forward classification; Alec Dixon: “the real quantitative quality marker in cannabis is terpene concentration”

2022 Jikomes et al. (PLOS ONE) Analysis of 89,923 commercial cannabis samples confirmed terpene profiles—not indica/sativa labels—are key chemical differentiators Empirical validation at population scale: traditional taxonomy is chemically meaningless; terpene profiles provide the only reliable chemical classification

2024 Schwarz et al. A2a mechanism (PAIN) Identified adenosine A2a as primary terpene analgesic mechanism; same receptor studied in aromatherapy (Park 2011) and forest medicine (Li phytoncide pathway) Mechanistic linchpin for reunification: single receptor mechanism connects cannabis, aromatherapy, and forest medicine terpene literatures

2024 Spindle et al. human trial (Drug Alcohol Depend.) First controlled human evidence: d-limonene selectively reduced THC-induced anxiety at 15 mg vaporized dose Translational bridge: demonstrates clinically meaningful terpene-cannabinoid interaction at realistic human doses ---

X. Conclusion: The Evidence Supports a Unified Framework with Critical Caveats

The evidence assembled in this review supports the conclusion that a unified terpene pharmacology—bridging aromatherapy, cannabis science, and forest medicine—is both

scientifically justified and practically achievable, but that critical unresolved questions must be addressed before the framework can be considered established.

The core pharmacological argument rests on five pillars. First, individual terpenes engage a multi-target receptor landscape extending far beyond CB1/CB2 to include adenosine A2a, GABA-A, TRP channels, serotonergic receptors, and cholinergic enzymes (Section 1). Second, inhaled terpenes achieve rapid brain penetration through at least three pathways—olfactory-limbic signaling, direct nose-to-brain transport, and respiratory-circulatory absorption—with pharmacokinetic parameters supporting therapeutic relevance at achievable exposure levels (Section 2). Third, aromatherapy RCTs encompassing 15,000–25,000 participants demonstrate clinically meaningful effects of inhaled terpenes, with the Silexan linalool program (n = 1,213 in double-blind RCTs) achieving anxiolysis equivalent to paroxetine and regulatory approval in Germany (Section 3). Fourth, the entourage effect has evolved from a theoretical concept to a mechanistically characterized phenomenon, with the A2a receptor identified as the primary terpene analgesic mechanism and the first controlled human evidence demonstrating selective limonene-mediated anxiolysis during THC administration (Section 4). Fifth, forest bathing immunology demonstrates that airborne terpenes produce measurable immune activation (NK cell effect size 2.50) and serotonergic/oxytocinergic effects through the same molecules studied in the cannabis and aromatherapy literatures (Section 5).

The molecular architecture supporting cross-pathway integration has been characterized with increasing precision. The A2a-CB1 heteromer system provides the physical substrate where terpene and cannabinoid signaling converge, with brain region–specific distribution explaining the selectivity of entourage effects (Section 6). Network pharmacology provides the intellectual framework for understanding why multi-target terpene activity represents an evolved design feature rather than a pharmacological limitation, and for designing cross-botanical terpene blends using mathematical optimization frameworks (Section 7).

Three critical unresolved issues warrant emphasis. First, the dose gap between animal studies (200 mg/kg IP) and human cannabis consumption (~0.01–0.07 mg/kg of any individual

terpene per session) remains the field's central translational challenge. The Raz et al. (2023) finding that terpene-mediated CB1 amplification occurs at concentrations as low as 0.001–0.01 μM partially addresses this gap for the cannabinoid synergy pathway, and the Spindle et al. (2024) demonstration that just 15 mg of vaporized limonene produced selective anxiolysis provides the strongest evidence that clinically meaningful effects can occur at realistic human doses. Nevertheless, the three-to-four orders of magnitude separating preclinical dosing from human terpene exposure during cannabis consumption has not been systematically resolved.

Second, the inhalation route did not produce terpene antinociception in the Schwarz et al. (2024) mouse model—a significant limitation for translational relevance to vaporized cannabis. Whether this reflects insufficient terpene delivery through the inhalation apparatus used, species differences in respiratory anatomy, or a genuine pharmacological constraint on terpene analgesia via inhalation remains undetermined. This negative finding coexists with the extensive positive evidence for inhaled terpene effects from the aromatherapy and forest medicine literatures, suggesting that the inhalation route may be effective for some endpoints (anxiolysis, cognition, immune modulation) but not others (peripheral nociception).

Third, the federal funding infrastructure that produced the field's most important advances faces existential threat from the proposed elimination of NCCIH in the FY2026 budget (Section 8). The convergence of NCCIH elimination proposals, NIDA contract cancellations, and uncertain cannabis rescheduling timelines creates a window of maximum vulnerability precisely when the field has achieved unprecedented mechanistic clarity.

The pre-loading paradigm—using essential oil–derived terpene exposure to prime receptor systems before cannabis consumption—represents the most direct translational application of the unified framework. With A2a-CB1 heteromers concentrated in striatopallidal neurons and hippocampal CA1 synapses, the molecular machinery for terpene-cannabinoid cross-talk is established. Whether the concentration gap between ambient terpene levels and pharmacologically active doses can be reliably bridged through olfactory signaling, prolonged low-dose exposure, or supplementation with essential oil inhalation remains the key empirical

question for this field's next decade. The answer will determine whether unified terpene pharmacology achieves the status of an experimentally validated paradigm or remains, as it is today, a convergent body of evidence awaiting its definitive integration.

Section 9 + Conclusion word count: ~3,600 words | Cumulative (Sections 1–10): ~25,500 words

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Year	**Event**	**Description**	**Significance for Disciplinary Fragmentation / Convergence**
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1982	Shinrin-yoku coined (Akiyama, Japanese MAFF)	Third discipline established; formal forest bathing research began with Li 2004	Added third parallel literature studying α-pinene and limonene as "phytoncides"—same molecules, third nomenclature, third set of journals

1993–2008	Essential oil industry growth (Young Living 1993, dōTERRA 2008)	Industry grew to $26.91B globally by 2024 (Polaris Market Research); dōTERRA alone: 5 million customers	Market forces commercialized terpenes outside cannabis regulatory framework; reinforced disciplinary divide by creating economic incentives for separation
2019–2020	Leafly abandons indica/sativa taxonomy	Cannabis guide restructured around cannabinoid ratios and 8 color-coded terpene profiles; Flower Finder tool searches by terpene preference	Most influential cannabis platform signals terpene primacy; implicit recognition that terpene profiles, not cultivar labels, determine effects
2021	Emerald Cup restructures judging by terpene	Six terpene-based genres (myrcene, β-caryophyllene, limonene, terpinolene, α-pinene, β-ocimene) replace indica/sativa; SC Labs partnership	Premium cannabis competition validates terpene-forward classification; Alec Dixon: “the real quantitative quality marker in cannabis is terpene concentration”
2022	Jikomes et al. (PLOS ONE)	Analysis of 89,923 commercial cannabis samples confirmed terpene profiles—not indica/sativa labels—are key chemical differentiators	Empirical validation at population scale: traditional taxonomy is chemically meaningless; terpene profiles provide the only reliable chemical classification
2024	Schwarz et al. A2a mechanism (PAIN)	Identified adenosine A2a as primary terpene analgesic mechanism; same receptor studied in aromatherapy (Park 2011) and forest medicine (Li phytoncide pathway)	Mechanistic linchpin for reunification: single receptor mechanism connects cannabis, aromatherapy, and forest medicine terpene literatures
2024	Spindle et al. human trial (Drug Alcohol Depend.)	First controlled human evidence: d-limonene selectively reduced THC-induced anxiety at 15 mg vaporized dose	Translational bridge: demonstrates clinically meaningful terpene-cannabinoid interaction at realistic human doses

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